

Bazele Circuitelor Electronice

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“Citește, Nile, și bagă în cap măcar și mai târziu”

Scolie la “*Răspunsuri către Talasie*” de Sf. Maxim Mărturisitorul, Filocalia, vol.3, editura Humanitas, București, 2017

Bazele Circuitelor Electronice

= curs facultative pentru secția de calculatoare =

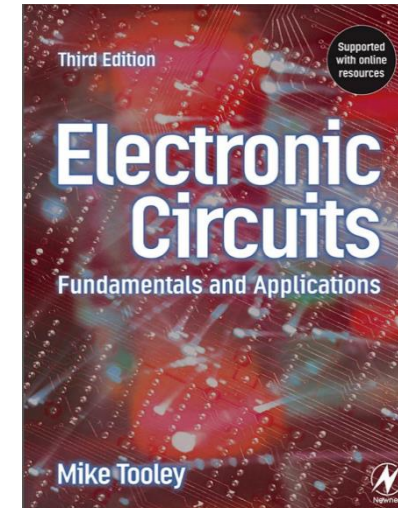
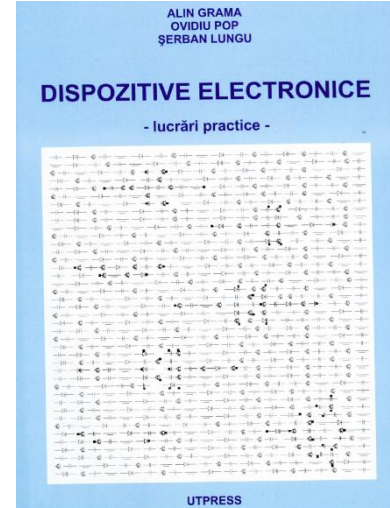
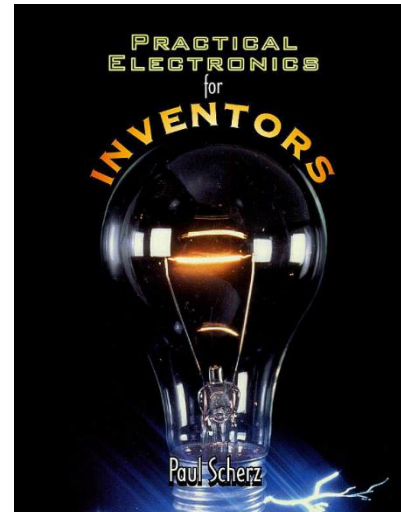
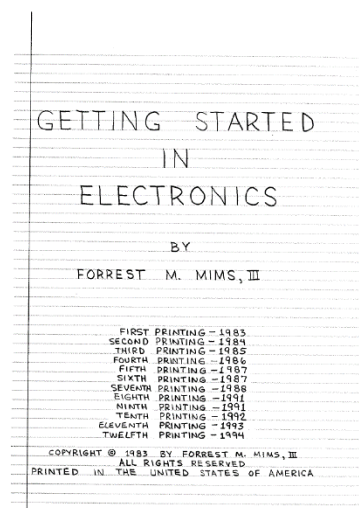
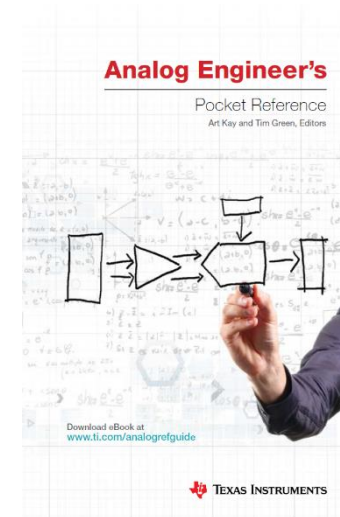
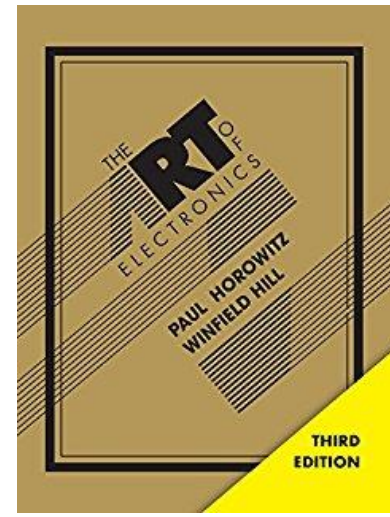
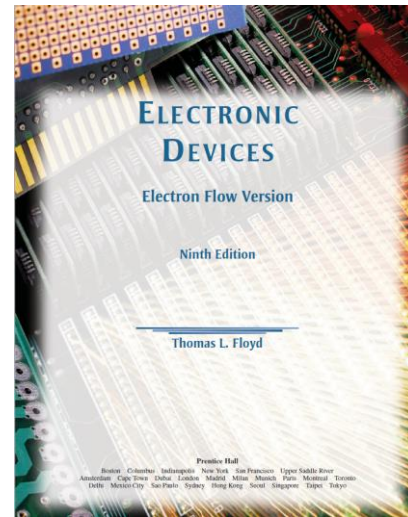
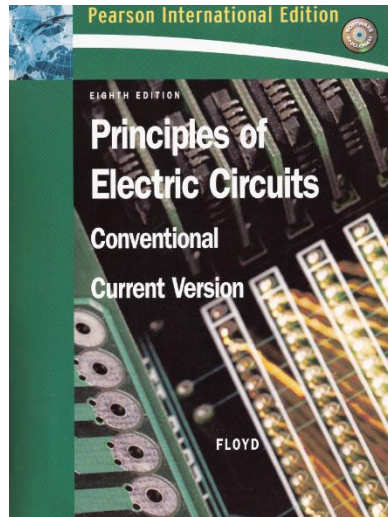
Personaje

- Curs
 - conf. dr. ing. Alin GRAMA
- Seminare
 - conf. dr. ing. Alin GRAMA
- Laboratoare
 - conf. dr. ing. Alin GRAMA
 - ing. Cristina DAVIDAȘ

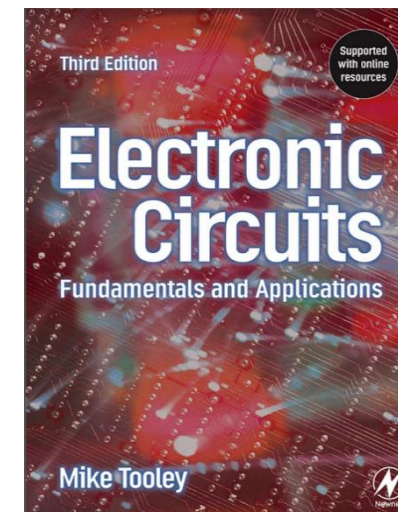
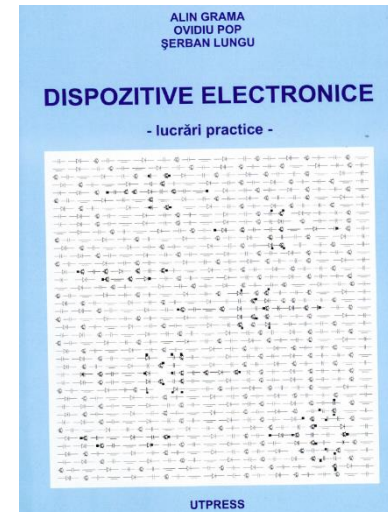
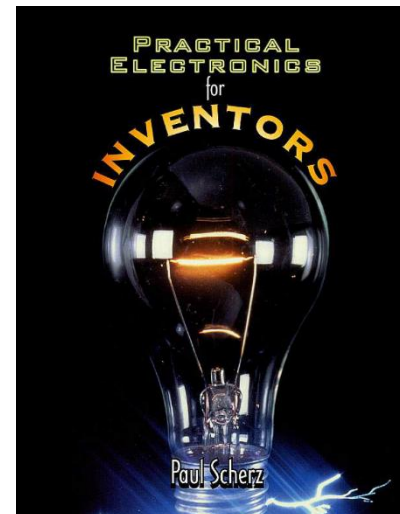
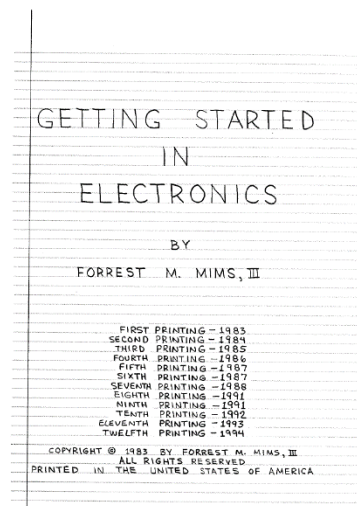
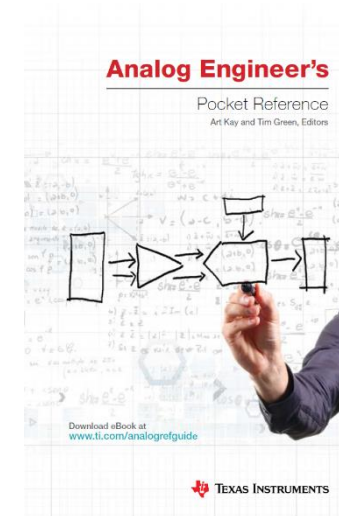
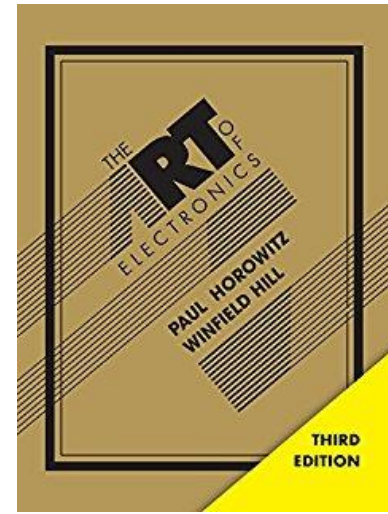
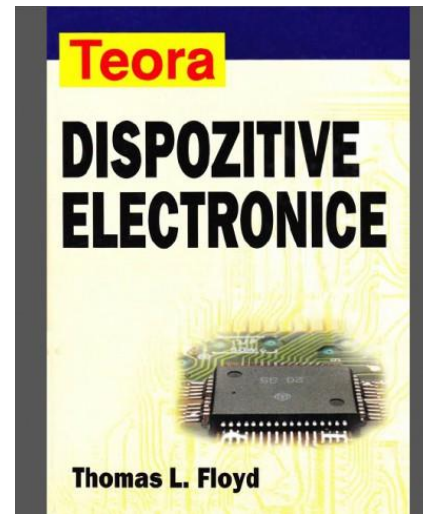
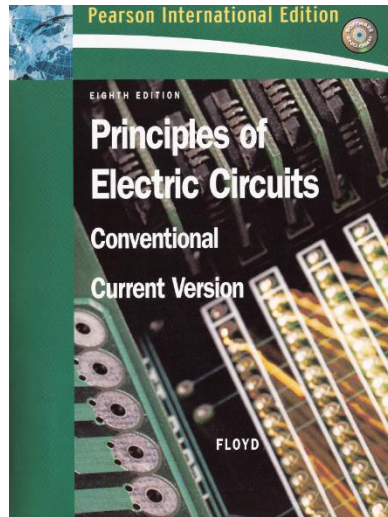
Nota

$$\text{Nota} = 80\% \text{Examen} + 20\% \text{Laborator}$$

Bibliografie



Bibliografie



Simulator

www.systemvision.com

The screenshot shows the SystemVision Cloud website. The header includes navigation links for Designs, Groups, Discuss, and Learn, along with a search bar and a Login button. The main banner features the text "Find. Design. Simulate. Collaborate." and a call to action "Choose parts, develop circuits, verify performance, communicate effectively". A large circuit diagram titled "Stepper Motor for Open-Loop Angle Control" is displayed, showing a motor driver circuit with a feedback loop. Below the banner, there are two featured articles: "Electro-Thermal Design Part 1: Digital Electronics" and "Electro-Thermal Design Part 2: Analog Electronics". The "What Interests You?" section lists various design categories with icons: Electro-Fluidic Systems and Control, Electro-Magnetic Components, Electronics Cooling and Thermal Control, Lighting Systems, Motors & Drives, Power Electronics, Power Systems, Renewable Energy, Robotics, Sensors, and Motion Control, and Signals and Communications. The footer contains the copyright notice "© 2020 SystemVision® Cloud All rights reserved." and a link to "Afiașă-le pe toate".

SystemVision® Cloud

Designs Groups Discuss Learn ?

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Find.
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Collaborate.

Choose parts, develop circuits, verify performance, communicate effectively

Create a Design > Find Groups >

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FEATURED ARTICLE
Electro-Thermal Design Part 1: Digital Electronics

FEATURED ARTICLE
Electro-Thermal Design Part 2: Analog Electronics

What Interests You?

Electro-Fluidic Systems and Control

Electro-Magnetic Components

Electronics Cooling and Thermal Control

Lighting Systems

Motors & Drives

Power Electronics

Power Systems

Renewable Energy

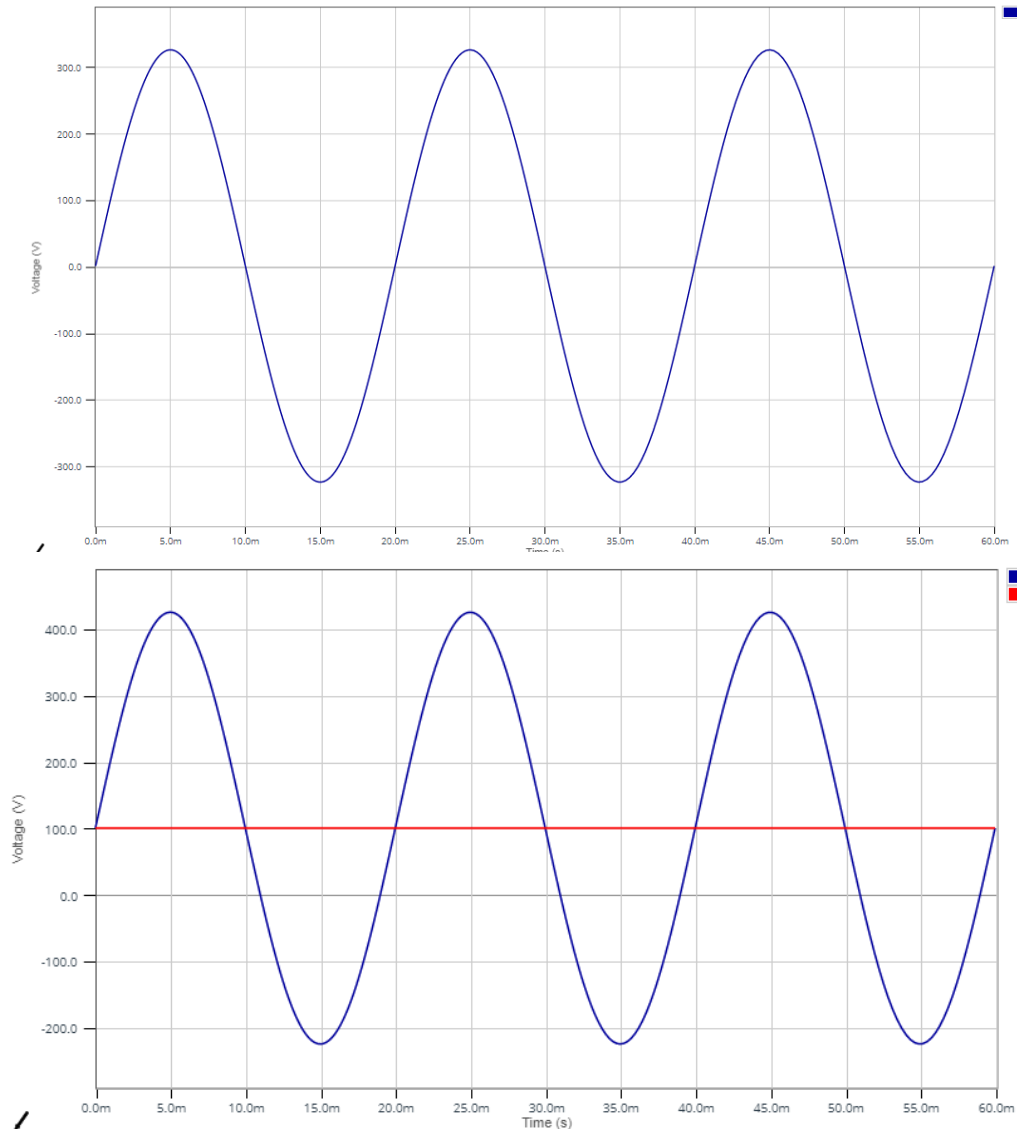
Robotics, Sensors, and Motion Control

Signals and Communications

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Afișăți-le pe toate X

Semnale



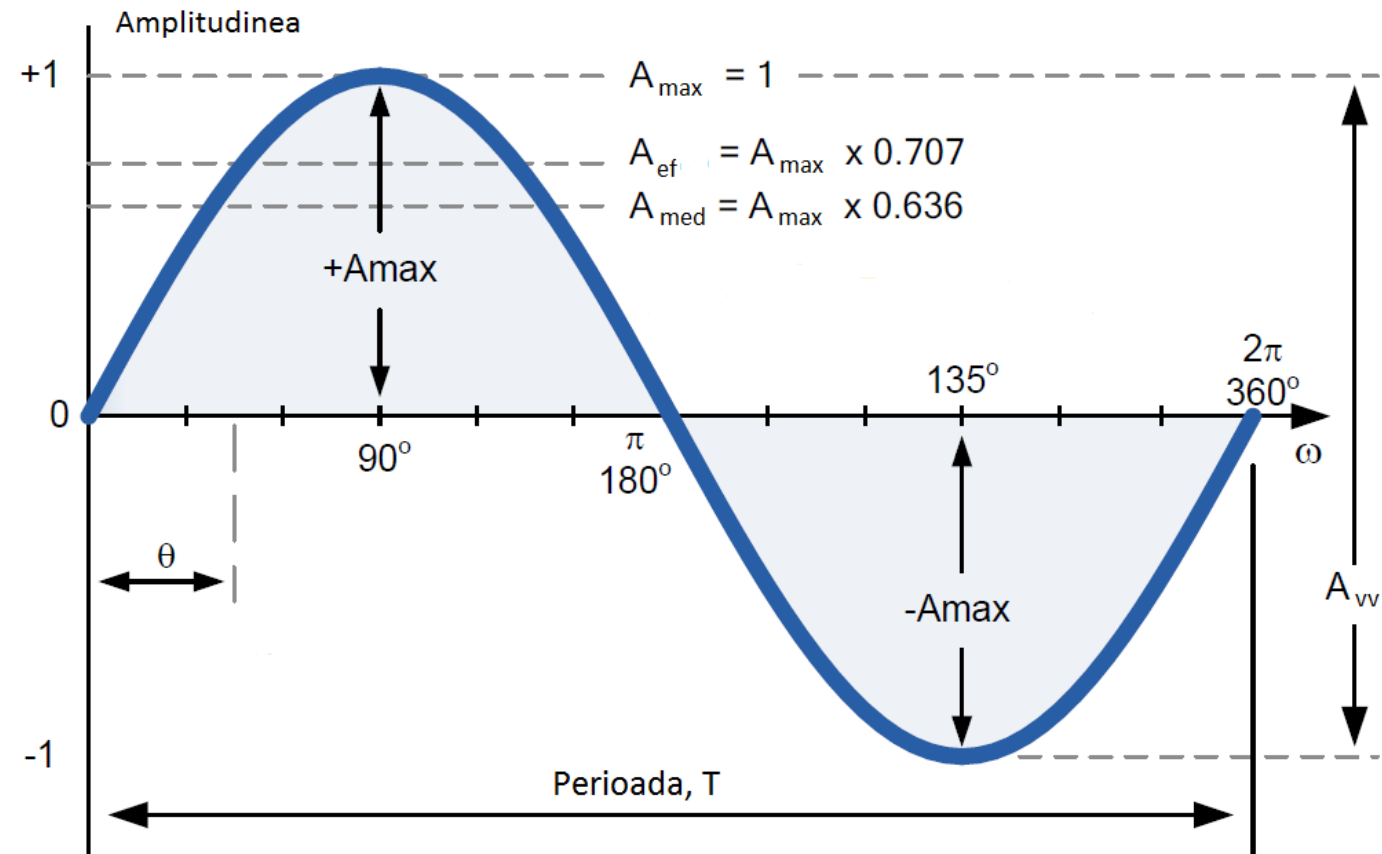
- A, f, T, phi

➤ V_{dc}

➤ V_{ef}

➤ V_{med}

Semnale

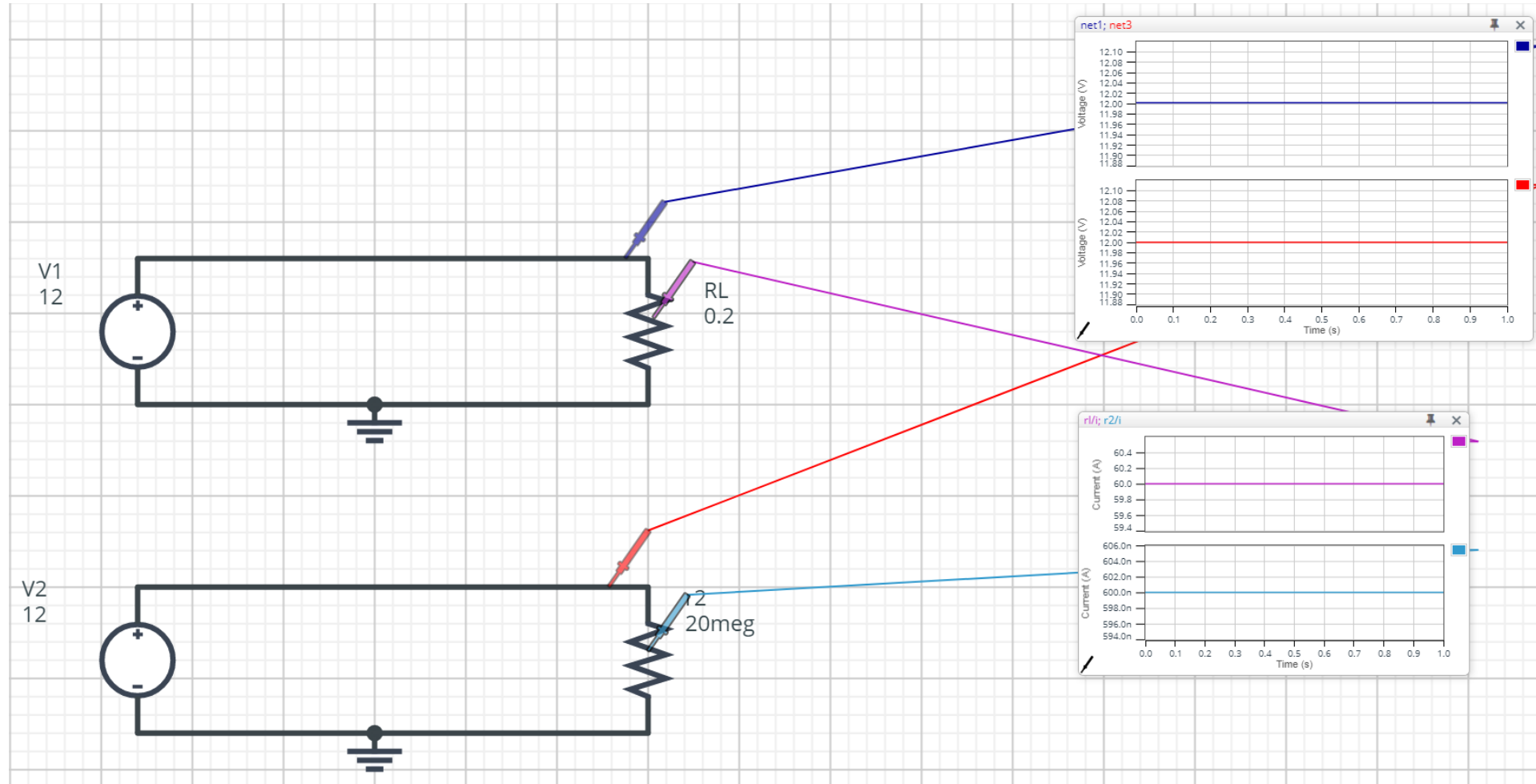


Surse de tensiune și de curent

- ideale
- reale

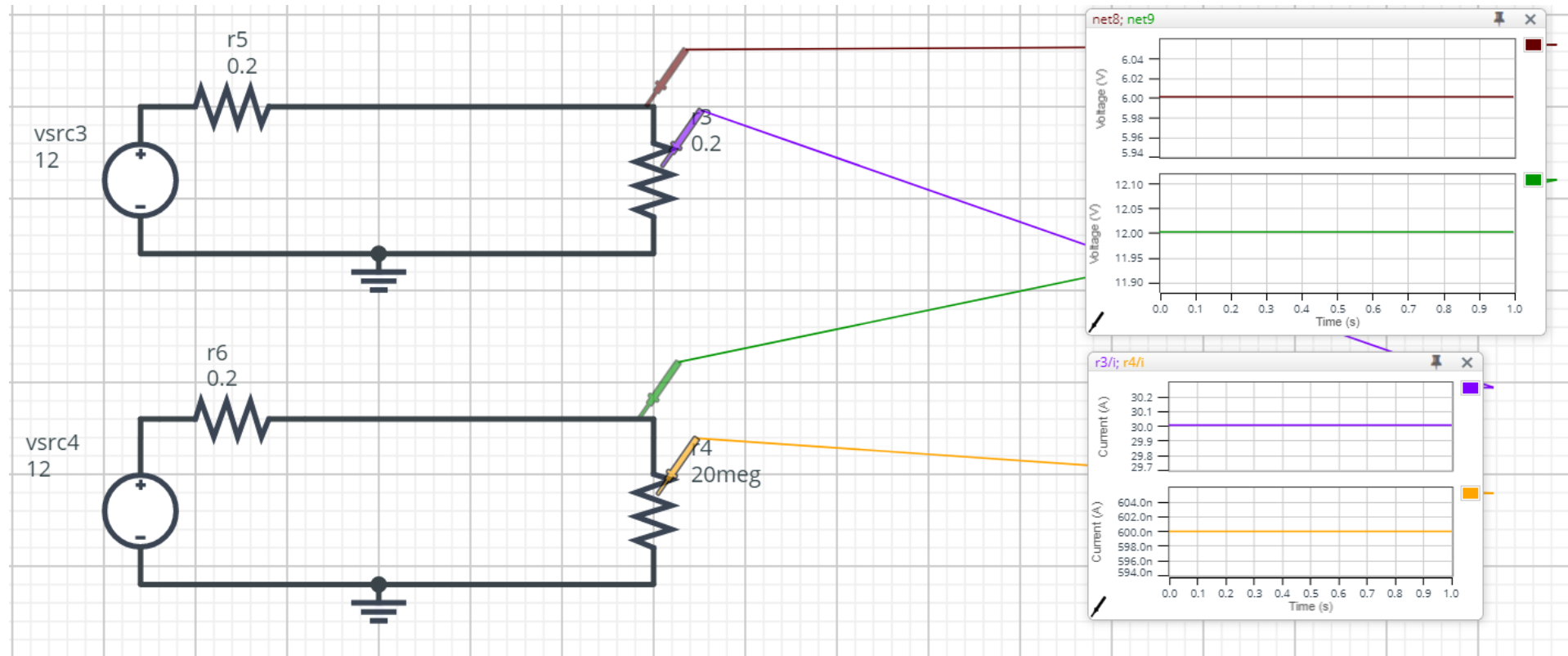
Surse ideale de tensiune

SV C Surse de tensiune - ideale

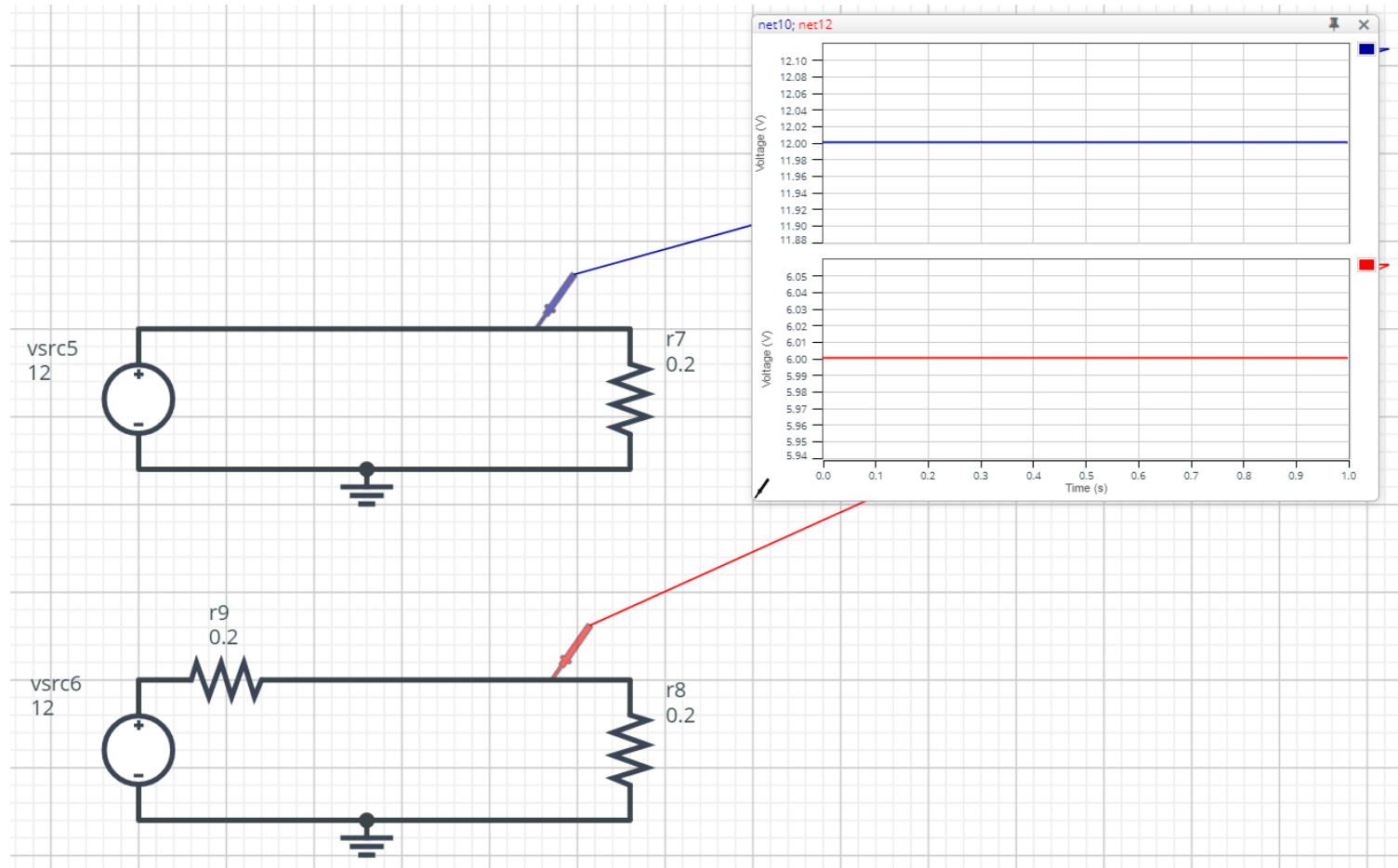


Surse reale de tensiune

SV
C Surse de tensiune - reale

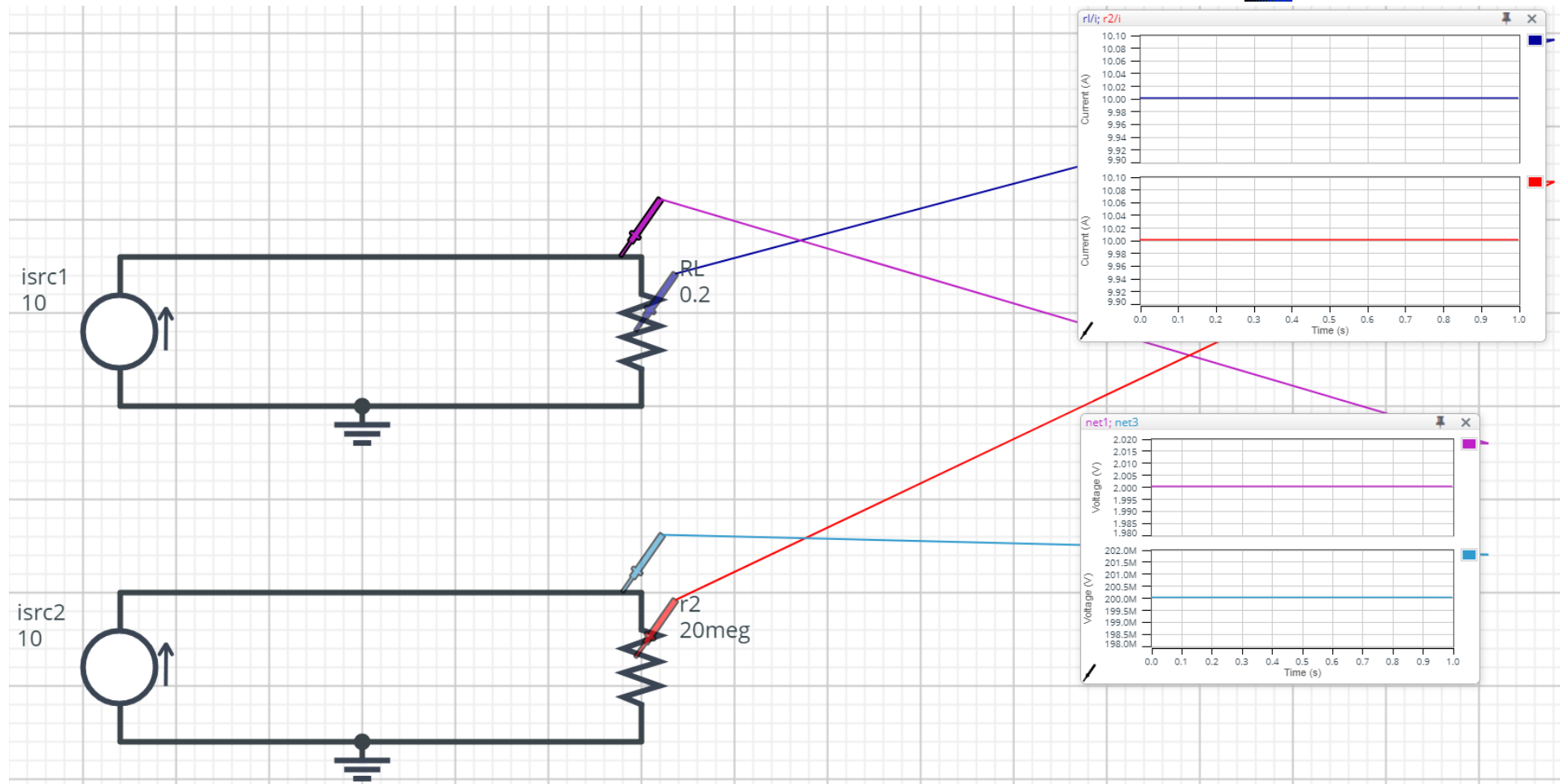


Surse de tensiune ideal vs real



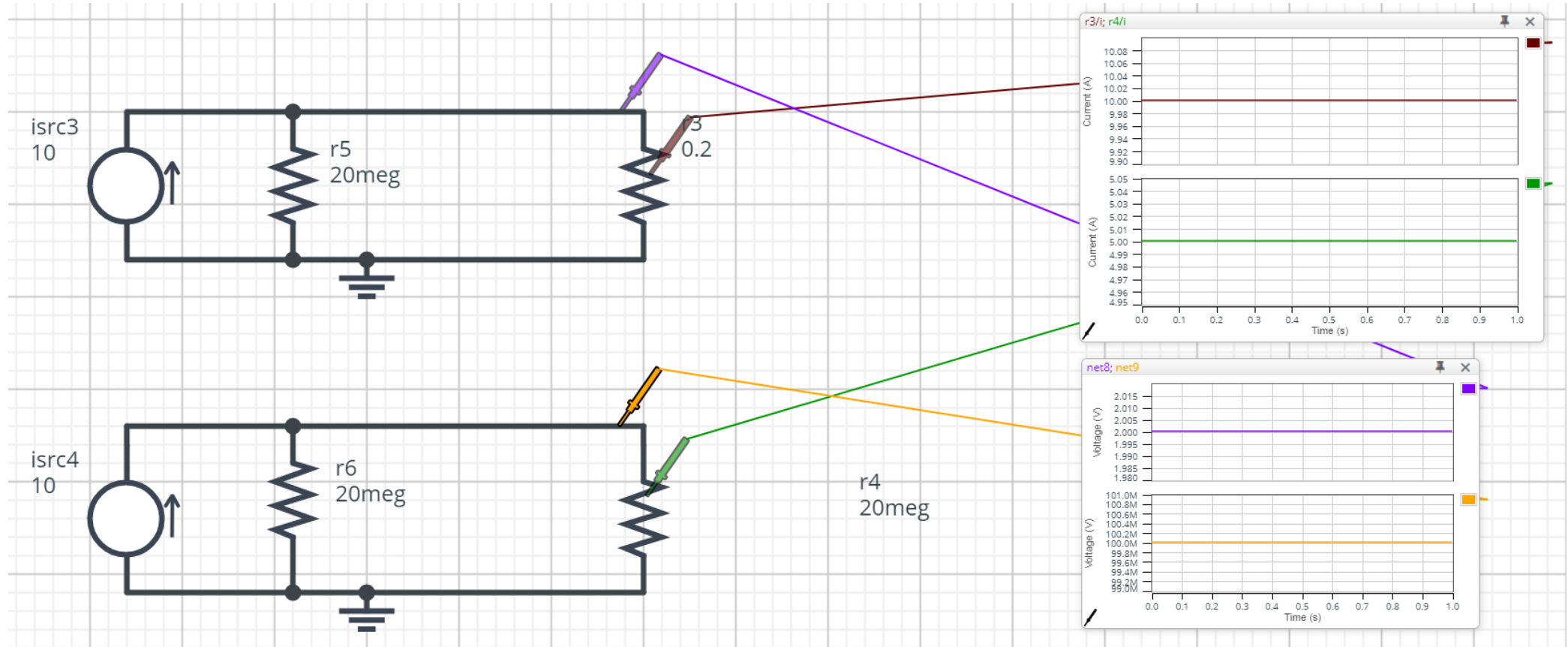
Surse ideale de curent

SV C Surse de curent - ideale/re

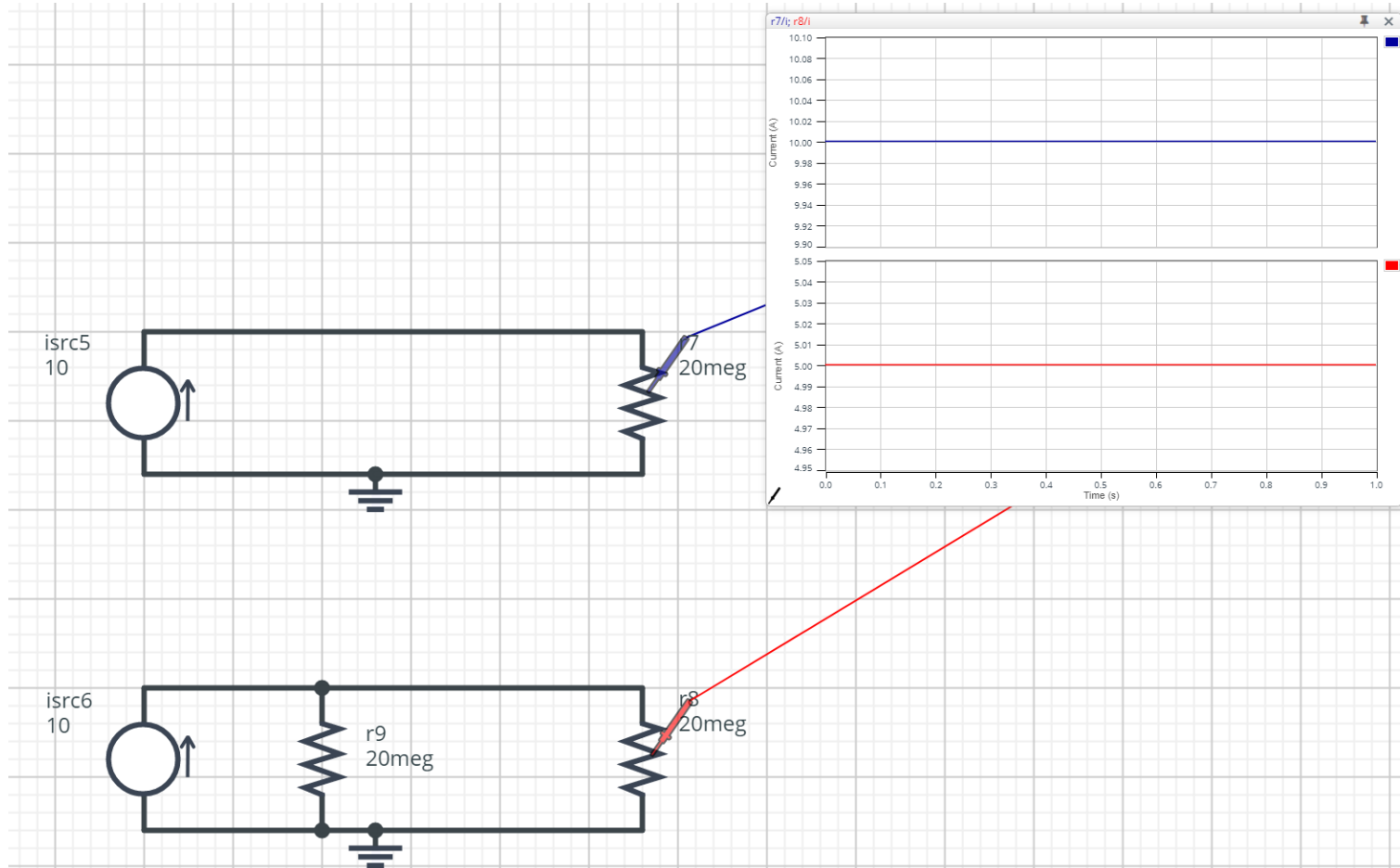


Surse reale de curent

SV
C Surse de curent - ideale/re

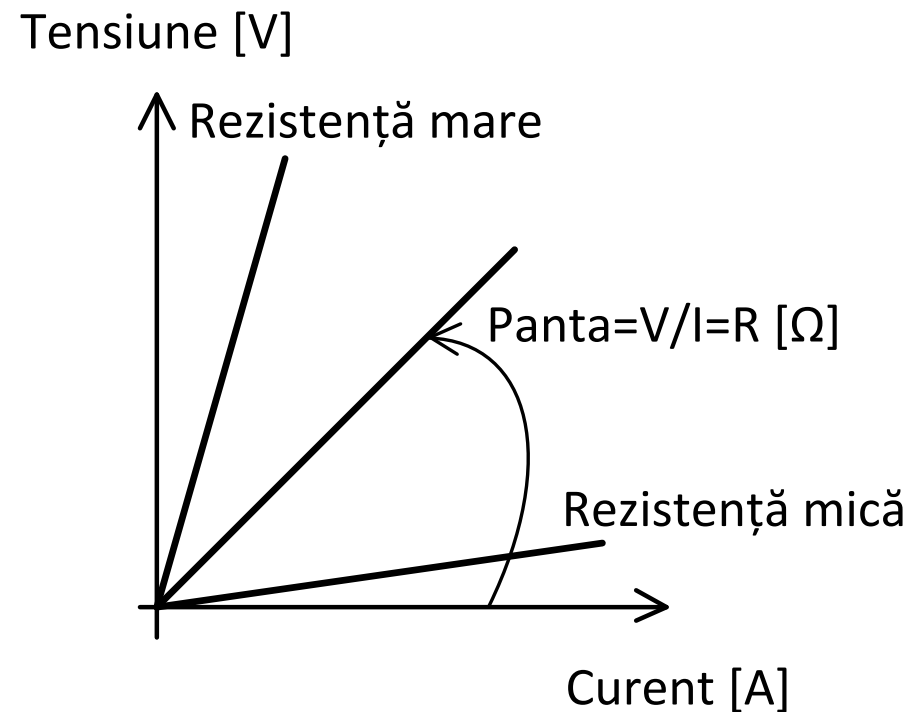


Surse de curent ideal vs real



Rezistența electrică

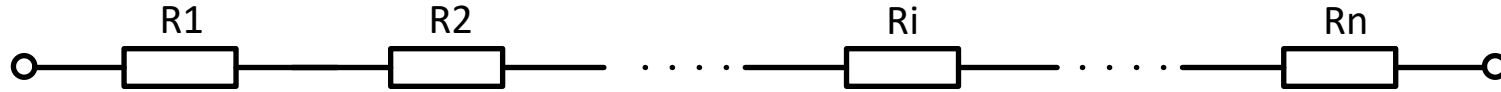
- Poate transforma curentul într-o cădere de tensiune și viceversa



Rezistența electrică

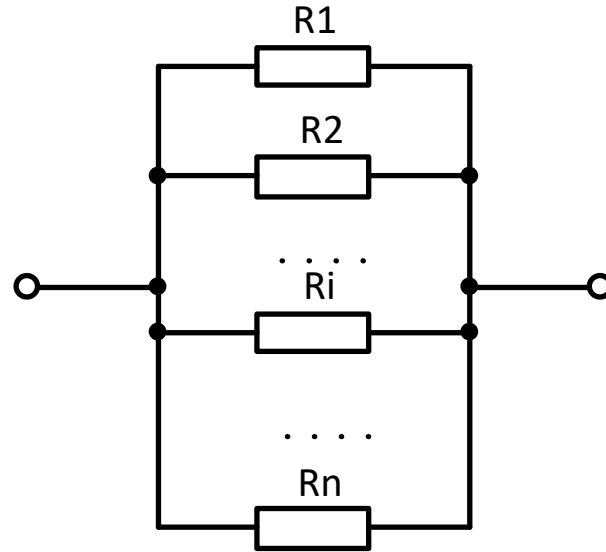
- La trecerea curentului electric printr-un material electronii liberi se ciocnesc de atomi ceea ce duce la pierderea de energie și, implicit, la îngreunarea deplasării.
- Def.: **Proprietatea unui material de a se opune trecerii curentului electric se numește Rezistență electrică.**

Rezistența electrică – conexiunea **serie**



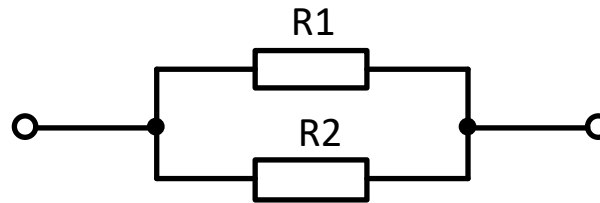
$$R_{echivalent} = \sum_{i=1}^n R_i$$

Rezistența electrică – conexiunea **paralel**



$$\frac{1}{R_{echivalent}} = \sum_{i=1}^n \frac{1}{R_i}$$

Rezistența electrică – conexiunea **paralel**

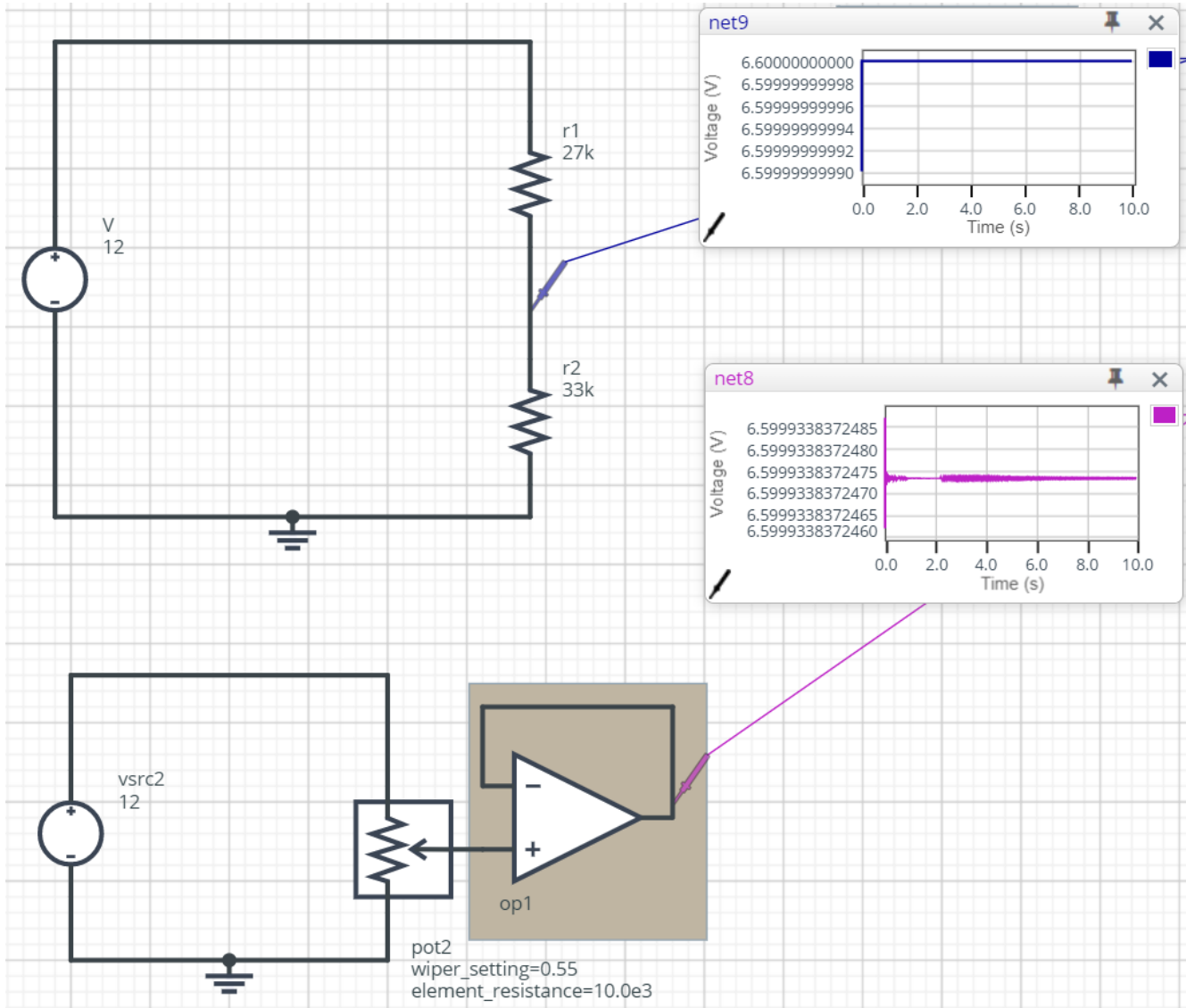


$$R_{echivalent} = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

Aplicații

- Divizorul de tensiune
- Divizorul de curent

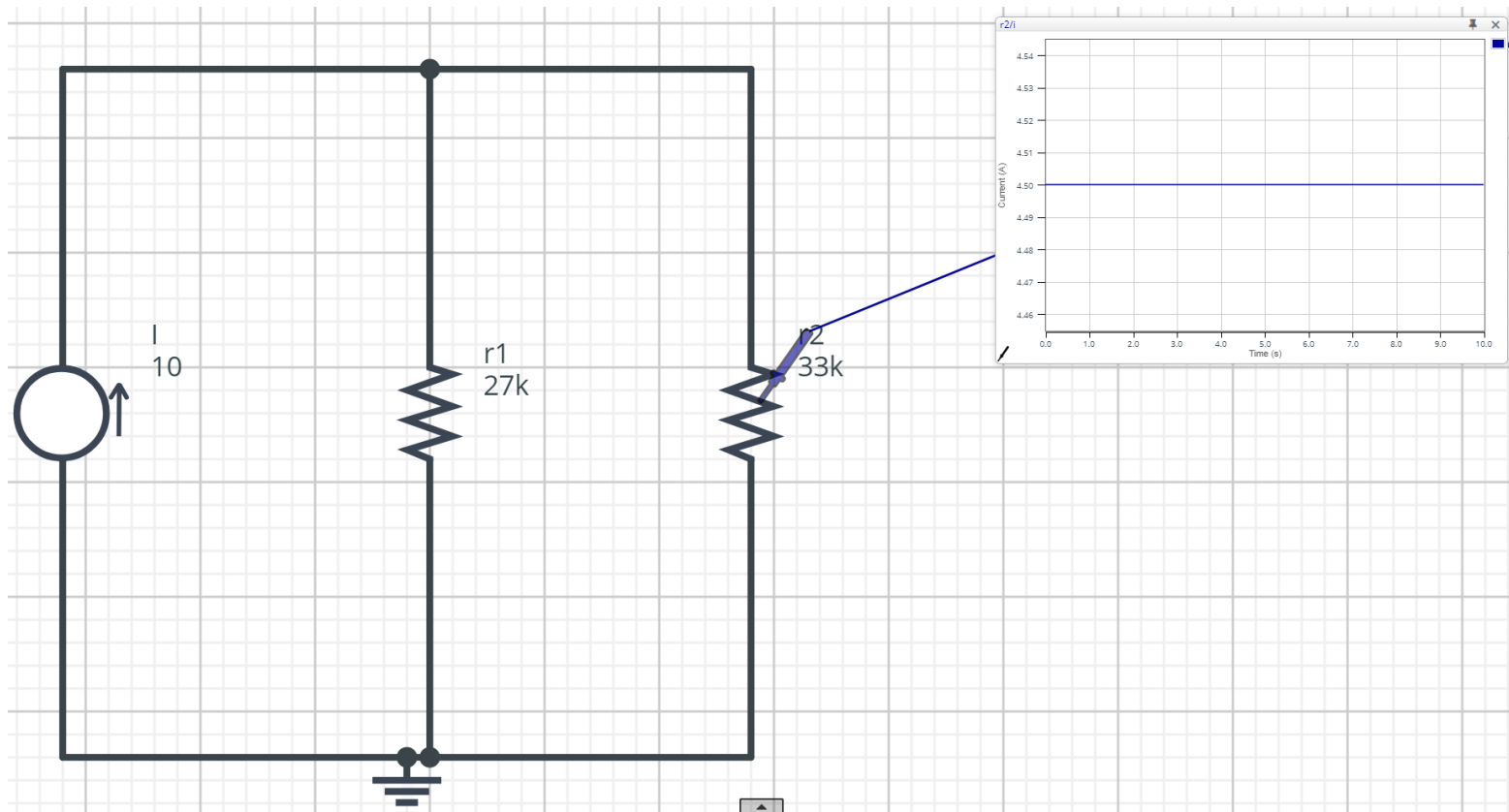
Divizorul de tensiune



$$V_{R2} = \frac{R_2}{R_1 + R_2} \cdot V$$

$$V_{R2} = \alpha \cdot V$$

Divizorul de curent



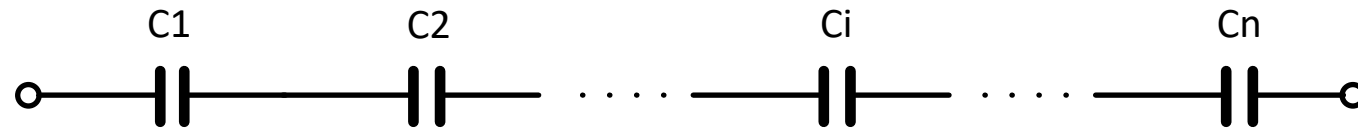
$$I_{R2} = \frac{R_1}{R_1 + R_2} \cdot I$$

SV
C Divizorul de curent

Capacitatea electrică

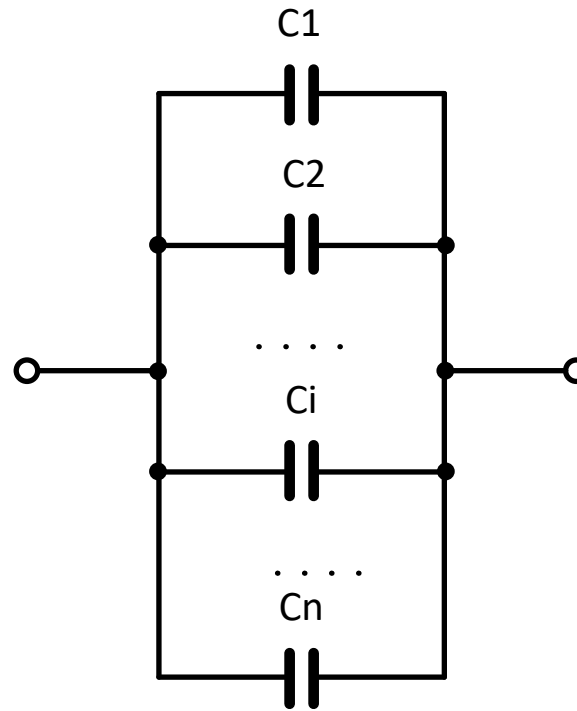
- Proprietatea de a înmagazina energie sub formă de câmp electric

Capacitatea – conexiunea **serie**



$$\frac{1}{C_{echivalent}} = \sum_{i=1}^n \frac{1}{C_i}$$

Capacitatea – conexiunea **paralel**

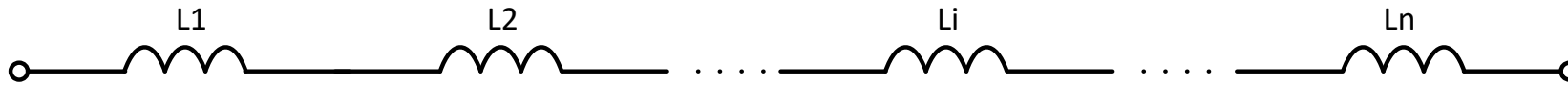


$$C_{echivalent} = \sum_{i=1}^n C_i$$

Inductanța

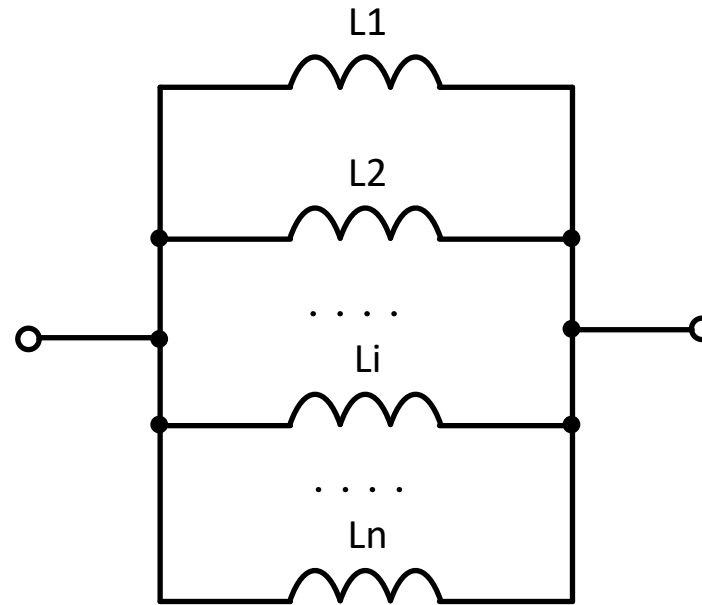
- Proprietatea de a înmagazina energie sub formă de câmp magnetic

Inductanța – conexiunea **serie**



$$L_{echivalent} = \sum_{i=1}^n L_i$$

Inductanța – conexiunea paralel



$$\frac{1}{L_{echivalent}} = \sum_{i=1}^n \frac{1}{L_i}$$

Componente pasive

- **NU** permit, numai prin funcționalitatea lor, transformarea energiei de curent continuu în energie de curent alternativ.
- Ca o consecință, cu ele nu se poate realiza amplificarea în putere a semnalelor variabile.
 - rezistorul
 - condensatorul
 - bobina

Componente active

- Permit transformarea energiei de curent continuu în energie de curent alternativ.
- Această proprietate permite să se obțină amplificarea semnalelor variabile în putere
 - tranzistoare
 - tiristoare
 - etc.

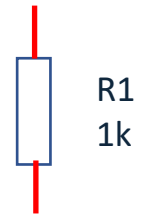
rezistorul

Parametri

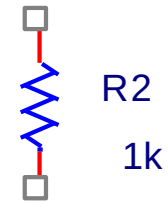
- Valoarea nominală
- Valoarea reală
- Toleranța
- Puterea maximă disipată
- Coeficientul de temperatură
- Zgomot
- Stabilitatea
- Temperatura ambientală

rezistorul

Simboluri



IEC



IEEE

rezistorul

Proprietăți

Proprietate	Carbon film	Metal film	Metal oxid	Ceramic	Sticlă	Metalizat RPM
Rezistența [Ω]	10 ÷ 10M	1 ÷ 1M	10 ÷ 10M	0,47 ÷ 22k	0,1 ÷ 22k	0,05 ÷ 10k
Toleranța [%]	± 5	± 1	± 2	± 5	± 5	± 5
Putere [W]	0,25 ÷ 2	0,125 ÷ 0,5	0,25 ÷ 0,5	4 ÷ 17	2 ÷ 4	10 ÷ 300
Coeficientul de temperatură [ppm/°C]	-250	+50 ÷ 100	+250	+250	+75	+50
Stabilitate	Acceptabilă	Excelentă	Excelentă	Bună	Bună	Bună
Temperatura mediului [°C]	-45 ÷ +125	-45 ÷ +125	-45 ÷ +125	-45 ÷ +125	-45 ÷ +125	-55 ÷ +200
Aplicații	General	Amplificatoare, echipamente de test etc.		Surse de alimentare, sarcini, electronică de putere		Electronică de putere ridicată

rezistorul

Material rezistive

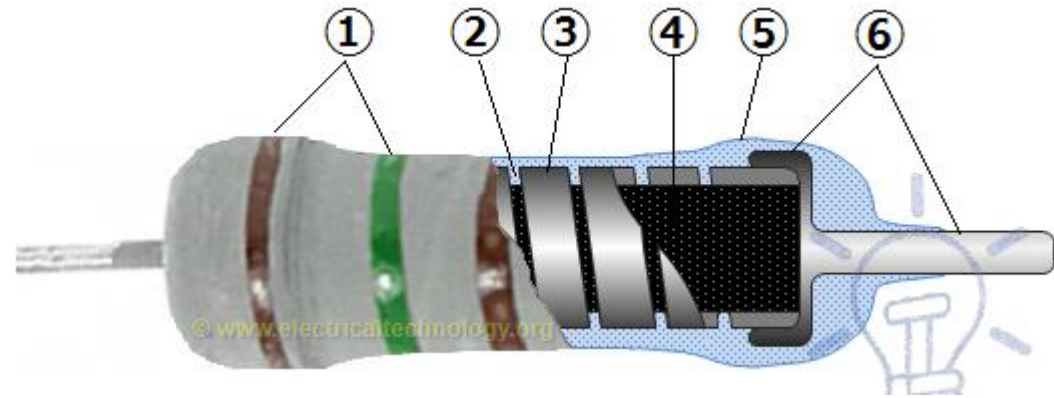
- **Carbon**



rezistorul

Material rezistive

- **Film de carbon**

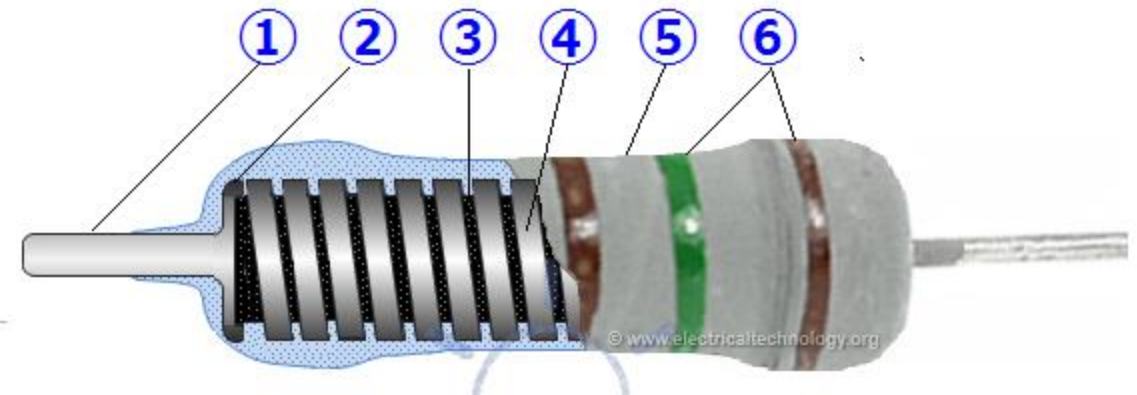


1	Benzi colorate
2	Spirală
3	Film de carbon
4	Substrat
5	Izolație
6	Terminale

rezistorul

Materiale rezistive

- **Film de metal**

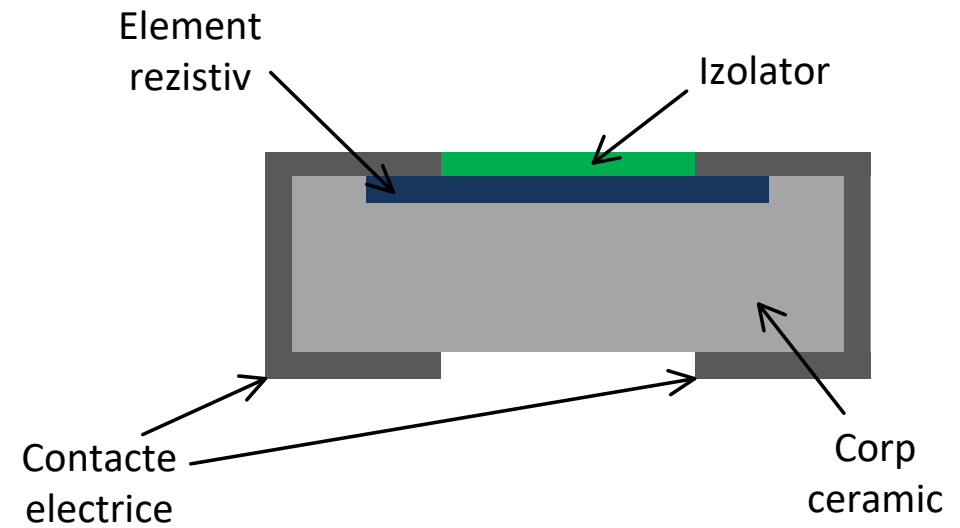


1	Terminale
2	Terminație
3	Substrat de ceramică
4	Film metalic
5	Izolație
6	Benzi colorate

rezistorul

SMD

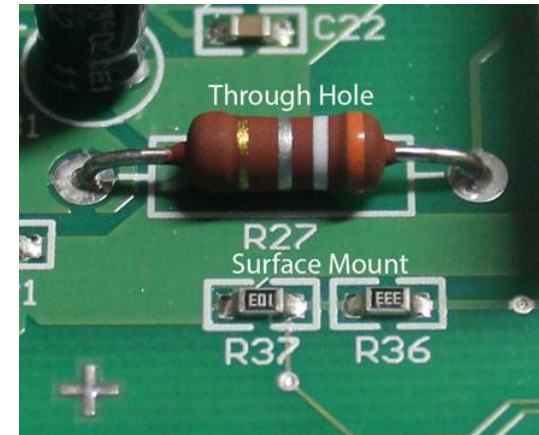
Surface Mount Device



rezistorul

SMD

Surface Mount Device



rezistorul

Valori
preferate

E6 (20%)	E12 (10%)	E24 (5%)	E48 (2%)	E96 (1%)	E192 (0.5%, 0.25%, 0.1%)	E6 (20%)	E12 (10%)	E24 (5%)	E48 (2%)	E96 (1%)	E192 (0.5%, 0.25%, 0.1%)	E6 (20%)	E12 (10%)	E24 (5%)	E48 (2%)	E96 (1%)	E192 (0.5%, 0.25%, 0.1%)
100	100	100	100	100	100	220	220	220	215	215	215	470	470	470	464	464	464
				101	101					218	218					470	470
				102	102					221	221					475	475
			105	104	104				226	223	223					481	481
				105	105					226	226					487	487
				106	106					229	229					493	493
		110	110	107	107			240	237	232	232			510	511	499	499
				109	109					234	234					505	505
				110	110					237	237					511	511
			113	111	111				243	240	240				536	517	517
				113	113					243	243					523	523
				114	114					246	246					530	530
		120	115	115	115			270	249	249	249			560	562	536	536
				117	117					252	252					542	542
				118	118					255	255					549	549
			121	120	120				261	258	258			620	619	556	556
				121	121					261	261					562	562
				123	123					264	264					569	569
150	120	120	124	124	124	270	270	270	267	267	267	680	680	680	681	576	576
				126	126					271	271					583	583
				127	127					274	274					590	590
			127	129	129				274	277	277				619	597	597
				130	130					280	280					604	604
				132	132					284	284					612	612
		130	133	133	133			300	287	287	287			750	750	619	619
				135	135					291	291					626	626
				137	137					294	294					634	634
			138	138	138				298	298	298				787	642	642
				140	140					301	301					649	649
				142	142					305	305					657	657
	150	150	140	143	143			330	301	309	309			820	825	665	665
				145	145					312	312					673	673
				147	147					316	316				866	681	681
		154	149	149	149			360	316	320	320					690	690
				150	150					324	324					706	706
				152	152					328	328					715	715
		160	154	154	154			390	332	332	332			910	909	715	715
				156	156					336	336					723	723
				158	158					340	340					732	732
			162	160	160				348	344	344				953	741	741
				162	162					348	348					750	750
				164	164					352	352					759	759
	180	180	165	165	165			430	357	357	357			953	953	768	768
				167	167					361	361					777	777
				169	169					365	365					787	787
		200	169	172	172			442	365	370	370				976	796	796
				174	174					374	374					806	806
				176	176					379	379					816	816
			178	178	178				383	383	383					825	825
				180	180					388	388					835	835
				182	182					392	392					845	845
		205	187	184	184			453	402	397	397				999	856	856
				187	187					402	402					866	866
				189	189					407	407					876	876
			191	191	191				412	412	412					887	887
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				196	196					422	422					909	909
	210	210	196	198	198			453	422	427	427				999	920	920
				200	200					432	432					931	931
				203	203					437	437					942	942
			205	205	205				442	442	442					953	953
				208	208					448	448					965	965
				210	210					453	453					976	976
				213	213					459	459					988	988

rezistorul

Valori preferate

IEC 60063														
E6 ±20%	E12 ±10%	E24 ±5%		E48 ±2%					E96 ±1%					
1.0	1.0 1.2	1.0	1.1	1.00	1.05	1.10	1.15		1.00	1.02	1.05	1.07	1.10	
		1.2	1.3	1.21	1.27	1.33	1.40	1.47	1.13	1.15	1.18	1.21	1.24	
									1.27	1.30	1.33	1.37	1.40	
									1.43	1.47				
1.5	1.5 1.8	1.5	1.6	1.54	1.62	1.69	1.78		1.50	1.54	1.58	1.62	1.65	
		1.8	2.0	1.87	1.96	2.05	2.15		1.69	1.74	1.78	1.82	1.87	
									1.91	2.00	2.05	2.10	2.15	
2.2	2.2 2.7	2.2	2.4	2.26	2.37	2.49	2.61		2.21	2.26	2.32	2.37	2.43	
		2.7	3.0	2.74	2.87	3.01	3.16		2.49	2.55	2.61	2.67	2.74	
									2.80	2.87	2.94	3.01	3.09	
									3.16	3.24				
3.3	3.3 3.9	3.3	3.6	3.32	3.48	3.65	3.83		3.32	3.40	3.48	3.57	3.65	
		3.9	4.3	4.02	4.22	4.42	4.64		3.74	3.83	3.92	4.02	4.12	
									4.22	4.32	4.42	4.53	4.64	
4.7	4.7 5.6	4.7	5.1	4.87	5.11	5.36			4.75	4.87	4.99	5.11	5.23	
		5.6	6.2	5.62	5.90	6.19	6.49		5.36	5.49	5.62	5.76	5.90	
									6.04	6.19	6.34	6.49	6.65	
6.8	6.8 8.2	6.8	7.5	6.81	7.15	7.50	7.87		6.81	6.98	7.15	7.32	7.50	
		8.2	9.1	8.25	8.66	9.09	9.53		7.68	7.87	8.06	8.25	8.45	
									8.66	8.87	9.09	9.31	9.53	
									9.76					

rezistorul

Valori preferate

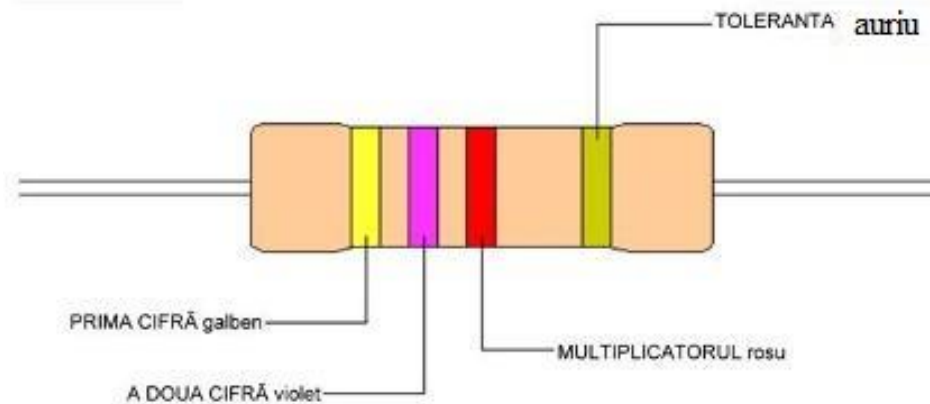
1.0	1.0	1.0
		1.1
	1.2	1.2
		1.3
1.5	1.5	1.5
		1.6
	1.8	1.8
		2.0
2.2	2.2	2.2
		2.4
	2.7	2.7
		3.0
3.3	3.3	3.3
		3.6
	3.9	3.9
		4.3
4.7	4.7	4.7
		5.1
	5.6	5.6
		6.2
6.8	6.8	6.8
		7.5
	8.2	8.2
E6		9.1
	E12	
		E24

Seria	Toleranța [%]
E6	± 20
E12	± 10
E24	± 5

rezistorul

Marcare

- Codul culorilor



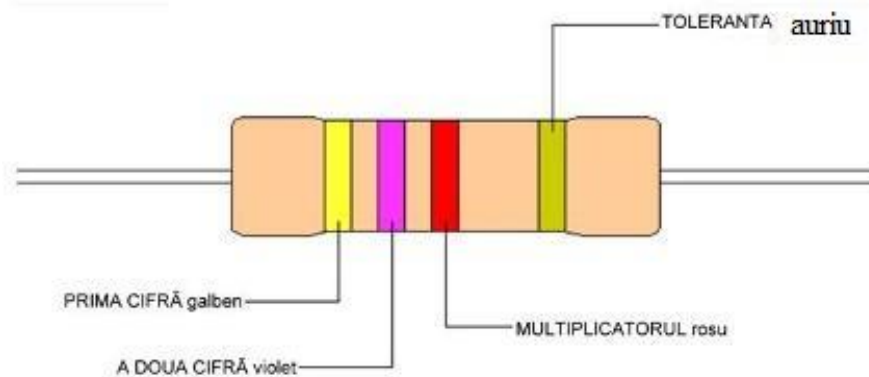
Numărul	Culoarea
0	negru
1	maro
2	roșu
3	portocaliu
4	galben
5	verde
6	albastru
7	violet
8	gri
9	alb

Toleranța	Culoarea
±1%	maro
±2%	roșu
±5%	auriu
±10%	argintiu

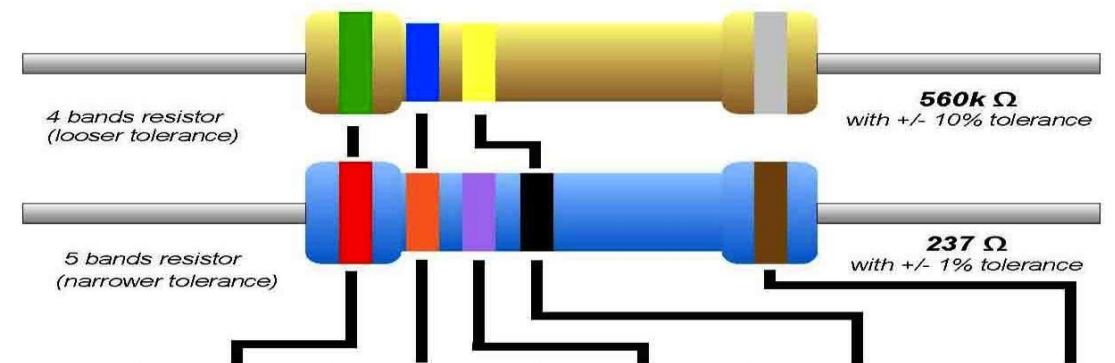
rezistorul

Marcare

- Codul culorilor



Resistor Color Code



Color	1 st Band	2 nd Band	3 rd Band	Multiplier	Tolerance
Black	0	0	0	$\times 1 \Omega$	
Brown	1	1	1	$\times 10 \Omega$	+/- 1%
Red	2	2	2	$\times 100 \Omega$	+/- 2%
Orange	3	3	3	$\times 1K \Omega$	
Yellow	4	4	4	$\times 10K \Omega$	
Green	5	5	5	$\times 100K \Omega$	+/- 5%
Blue	6	6	6	$\times 1M \Omega$	+/- .25%
Violet	7	7	7	$\times 10M \Omega$	+/- .1%
Grey	8	8	8		+/- .05%
White	9	9	9		
Gold				$\times .1 \Omega$	+/- 5%
Silver				$\times .01 \Omega$	+/- 10%

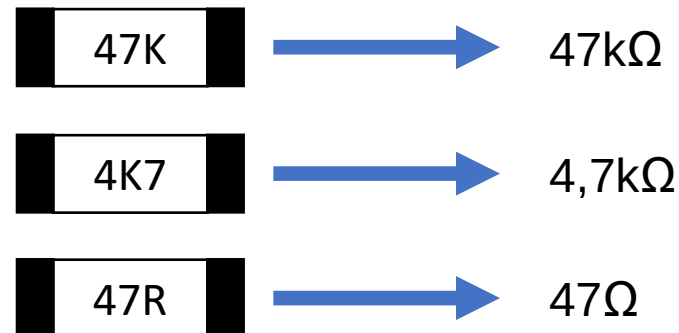
rezistorul

Marcare

- BS 1852 (BS EN 60062)

Litera	Multiplicator
R	1
K	10^3
M	10^6

Litera	Toleranța
F	$\pm 1\%$
G	$\pm 2\%$
J	$\pm 5\%$
K	$\pm 10\%$
M	$\pm 20\%$



rezistorul

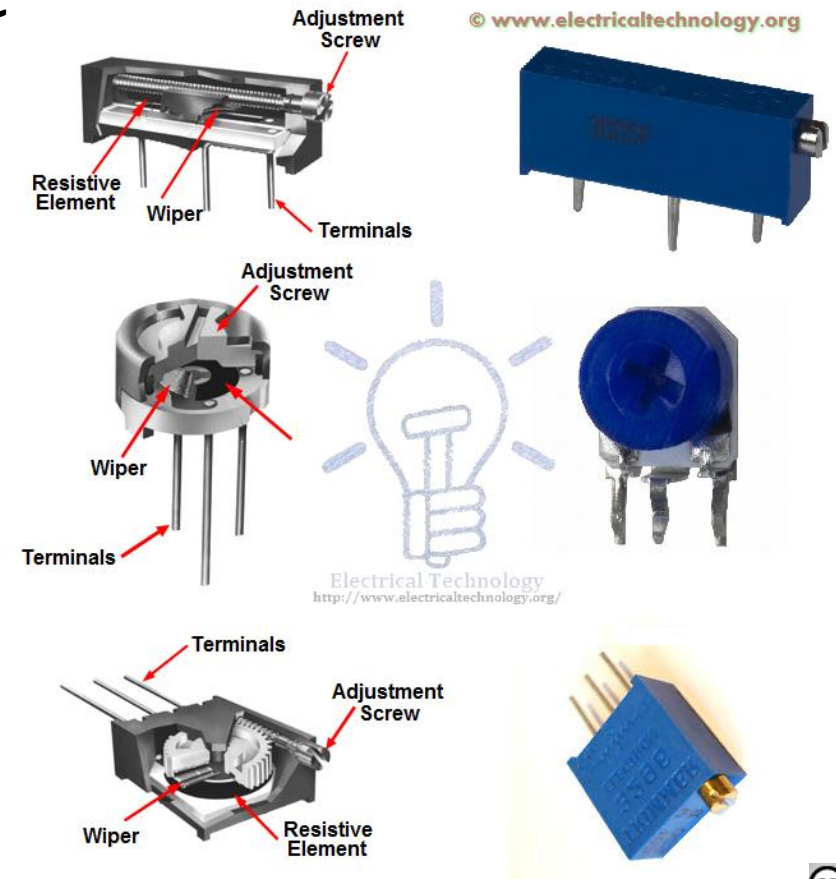
Variable

- Potențiometre



Variable

- Semireglabile



rezistorul

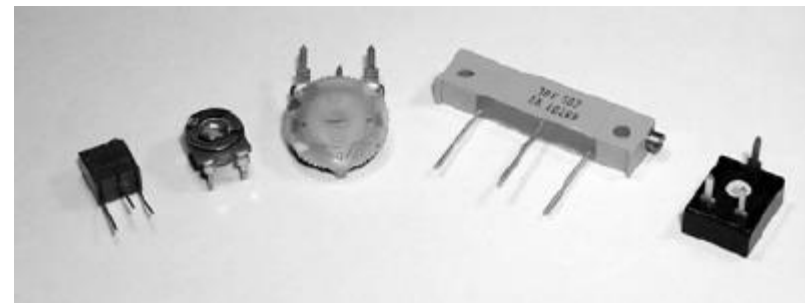
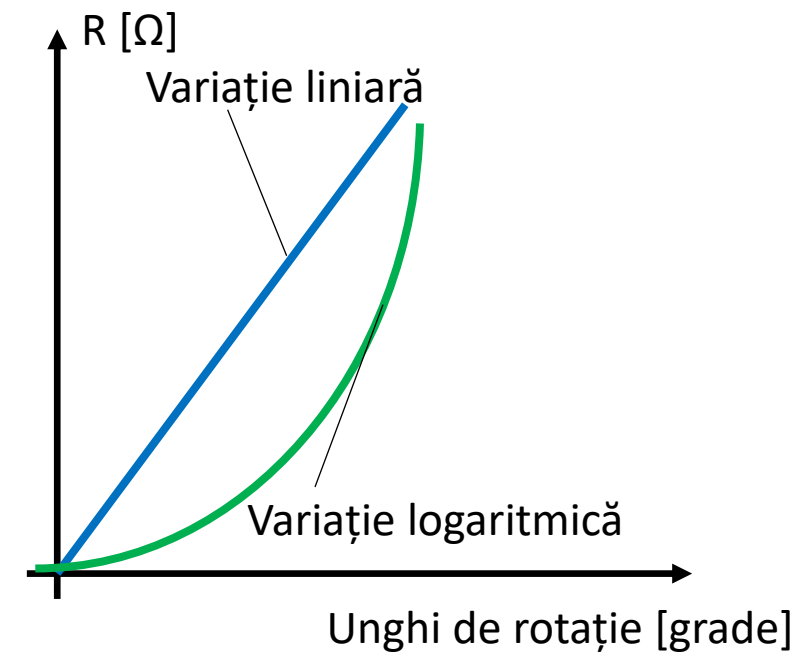
Variabile

- Potențiometre



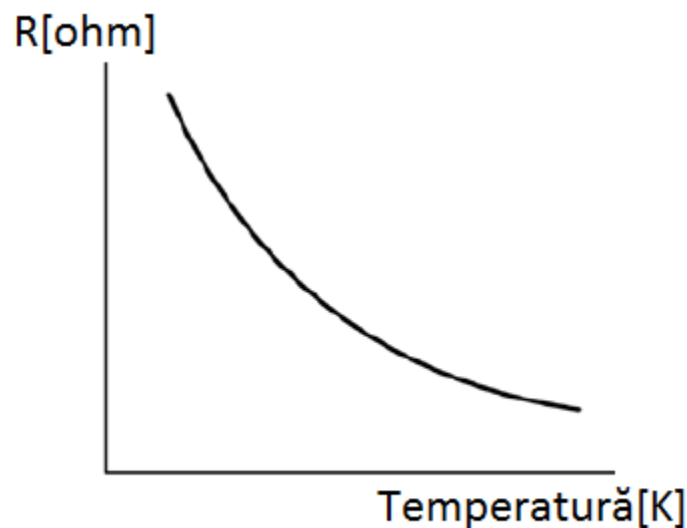
Variabile

- Semireglabile

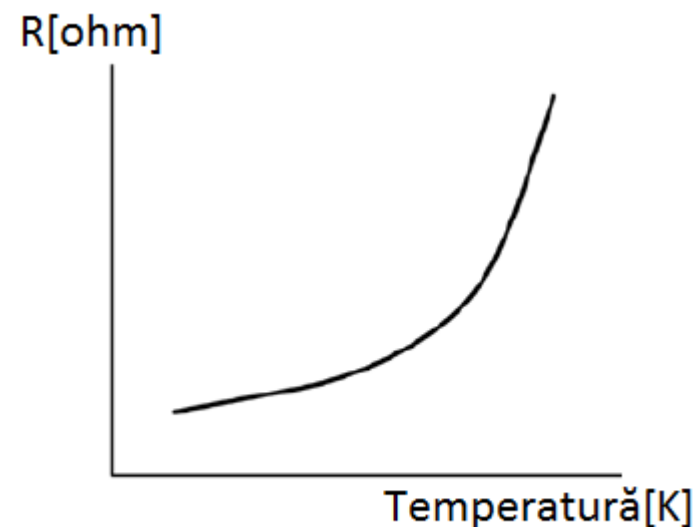


Termistorul

Rezistența la borne este dependentă de temperatură



(a) Negative temperature coefficient (NTC)



(b) Positive temperature coefficient (PTC)

$$R_T = R_0 \cdot e^{\beta \left(\frac{T_0 - T}{T_0 \cdot T} \right)}$$

Termistorul



www.vishay.com

NTCLE100E3

Vishay BCcomponents

NTC Thermistors, Radial Leaded, Standard Precision



FEATURES

- Accuracy over a wide temperature range
- High stability over a long life
- Excellent price/performance ratio
- UL recognized, file E148885
- Material categorization:
For definitions of compliance please see
www.vishay.com/doc/799912



RoHS
COMPLIANT

APPLICATIONS

- Temperature measurement, sensing and control, temperature compensation in industrial and consumer electronics

DESCRIPTION

These thermistors have a negative temperature coefficient. The device consists of a chip with two solid copper tin plated leads. It is grey lacquered and color coded, but not insulated.

PACKAGING

The thermistors are packed in bulk or tape on reel; see code numbers and relevant packaging quantities.

DESIGN-IN SUPPORT

For complete Curve Computation, visit:
www.vishay.com/resistors-non-linear/cure-computation-list/

MARKING

The thermistors are marked with colored bands; see dimensions drawing and "Electrical data and ordering information".

MOUNTING

By soldering in any position.
Not intended for potted applications.

ELECTRICAL DATA AND ORDERING INFORMATION								
R ₂₅ (Ω)	B _{25/85} -VALUE (K)	(± %)	UL APPROVED (Y/N)	SAP MATERIAL NUMBER NTCLE100E3...B0/T1/T2 (2)	OLD 12NC CODE 2381 640 3/4/6... (1)	COLOR CODE (3)		
						I	II	III
3.3	2880	3	N	338/B0	*338	Orange	Orange	Gold
4.7	2880	3	N	478/B0	*478	Yellow	Violet	Gold
6.8	2880	3	N	688/B0	*688	Blue	Grey	Gold
10	2990	3	N	109/B0	*109	Brown	Black	Black
15	3041	3	N	159/B0	*159	Brown	Green	Black
22	3136	3	N	229/B0	*229	Red	Red	Black
33	3390	3	Y	339/B0	*339	Orange	Orange	Black
47	3390	3	Y	479/B0	*479	Yellow	Violet	Black
68	3390	3	Y	689/B0	*689	Blue	Grey	Black
100	3560	1.5	Y	101/B0	*101	Brown	Black	Brown
150	3560	1.5	Y	151/B0	*151	Brown	Green	Brown
220	3560	1.5	Y	221/B0	*221	Red	Red	Brown
330	3560	1.5	Y	331/B0	*331	Orange	Orange	Brown

Revision: 24-Aug-12

Document Number: 29049

For technical questions, contact: nlb@vishay.com
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www.vishay.com

NTCLE100E3

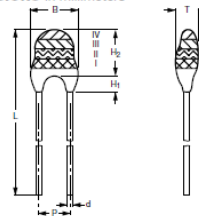
Vishay BCcomponents

ELECTRICAL DATA AND ORDERING INFORMATION								
R ₂₅ (Ω)	B _{25/85} -VALUE (K)	(± %)	UL APPROVED (Y/N)	SAP MATERIAL NUMBER NTCLE100E3...B0/T1/T2 ⁽²⁾	OLD 12NC CODE 2381 640 3/4/6... ⁽¹⁾	COLOR CODE ⁽³⁾		
						I	II	III
470	3560	1.5	Y	471"B0	*471	Yellow	Violet	Brown
680	3560	1.5	Y	681"B0	*681	Blue	Grey	Brown
1000	3528	0.5	Y	102"B0	*102	Brown	Black	Red
1500	3528	0.5	Y	152"B0	*152	Brown	Green	Red
2000	3528	0.5	Y	202"B0	*202	Red	Black	Red
2700	3977	0.75	Y	272"B0	*272	Red	Red	Red
3300	3977	0.75	Y	332"B0	*332	Orange	Orange	Red
4700	3977	0.75	Y	472"B0	*472	Yellow	Violet	Red
5000	3977	0.75	Y	502"B0	*502	Green	Black	Red
6800	3977	0.75	Y	682"B0	*682	Blue	Grey	Red
10 000	3977	0.75	Y	103"B0	*103	Brown	Black	Orange
12 000	3740	2	Y	123"B0	*123	Brown	Red	Orange
15 000	3740	2	Y	153"B0	*153	Brown	Green	Orange
22 000	3740	2	Y	223"B0	*223	Red	Red	Orange
33 000	4090	1.5	Y	333"B0	*333	Orange	Orange	Orange
47 000	4090	1.5	Y	473"B0	*473	Yellow	Violet	Orange
50 000	4190	1.5	Y	503"B0	*503	Green	Black	Orange
68 000	4190	1.5	Y	683"B0	*683	Blue	Grey	Orange
100 000	4190	1.5	Y	104"B0	*104	Brown	Black	Yellow
150 000	4370	2.5	Y	154"B0	*154	Brown	Green	Yellow
220 000	4370	2.5	Y	224"B0	*224	Red	Red	Yellow
330 000	4570	1.5	N	334"B0	*334	Orange	Orange	Yellow
470 000	4570	1.5	N	474"B0	*474	Yellow	Violet	Yellow

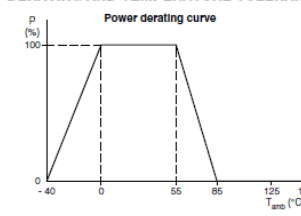
Notes

- (1) Replace * in 12NC by 3 for 5 %, 6 for 3 %, 4 for 2 %
- (2) Replace * in SAP by J for 5 %, H for 3 %, G for 2 %
- (3) For R₂₅ ± 2 % band IV is red, ± 3 % band IV is orange, ± 5 % band IV is gold

DIMENSIONS in millimeters



DERATING AND TEMPERATURE TOLERANCES



Note
• Zero power is considered as measuring power max. 1 % of max. power.

PHYSICAL DIMENSIONS FOR RELEVANT TYPE (all dimensions in millimeters)							
R ₂₅ -VALUE	B _{MAX}	d	H ₁		H ₂ MAX.	L	P
			MIN.	MAX.			
3.3 Ω to 220 Ω	5.0	0.6 ± 0.06	1.0	4.0	6.0	24 ± 1.5	2.54
330 Ω to 470 kΩ	3.3 ± 0.5	0.6 ± 0.06	1.0	3.0	6.0	24 ± 1.5	2.54

Revision: 24-Aug-12

2

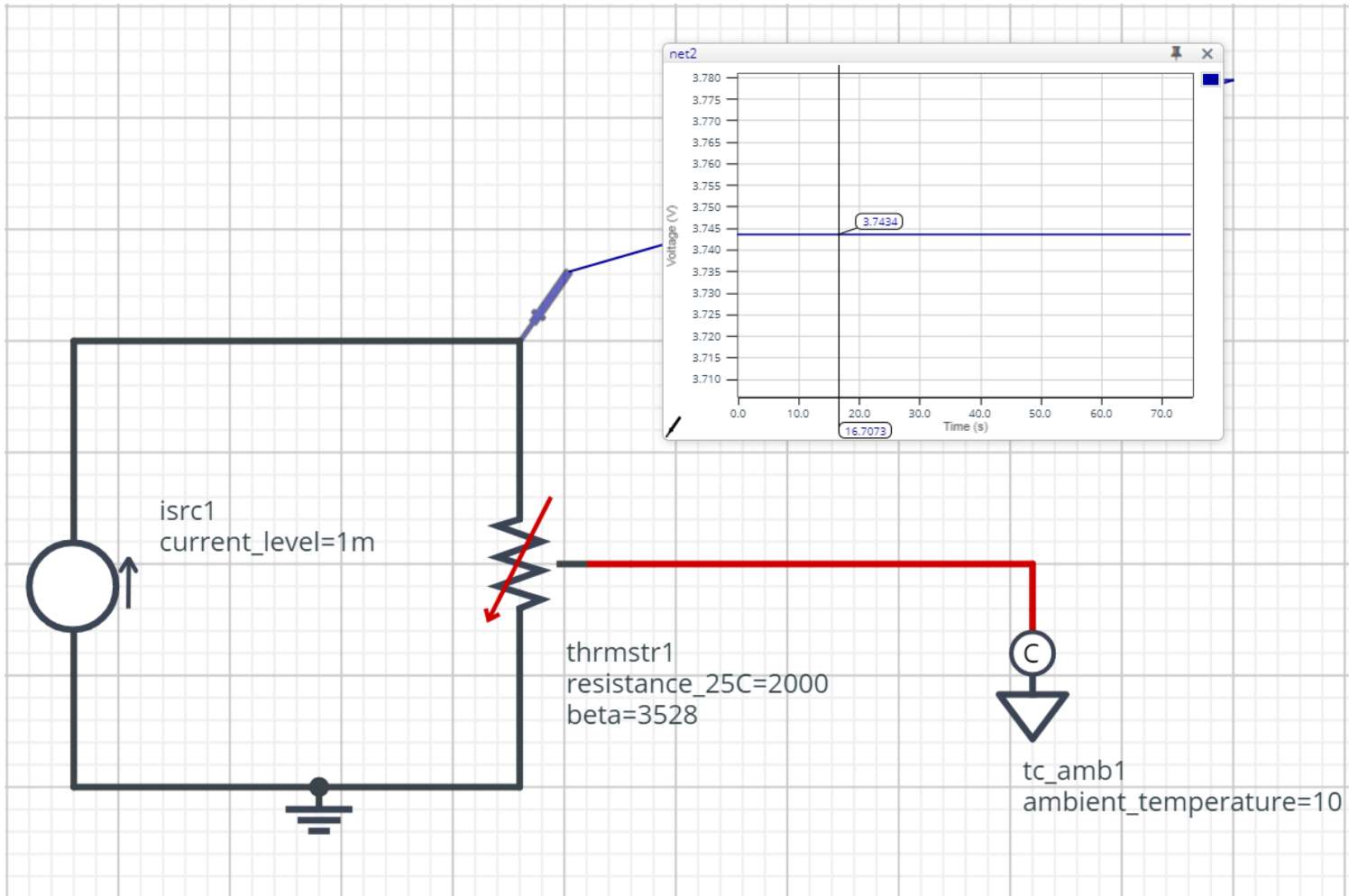
Document Number: 29049

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$$R_T = R_0 \cdot e^{\beta \left(\frac{T_0 - T}{T_0 \cdot T} \right)}$$

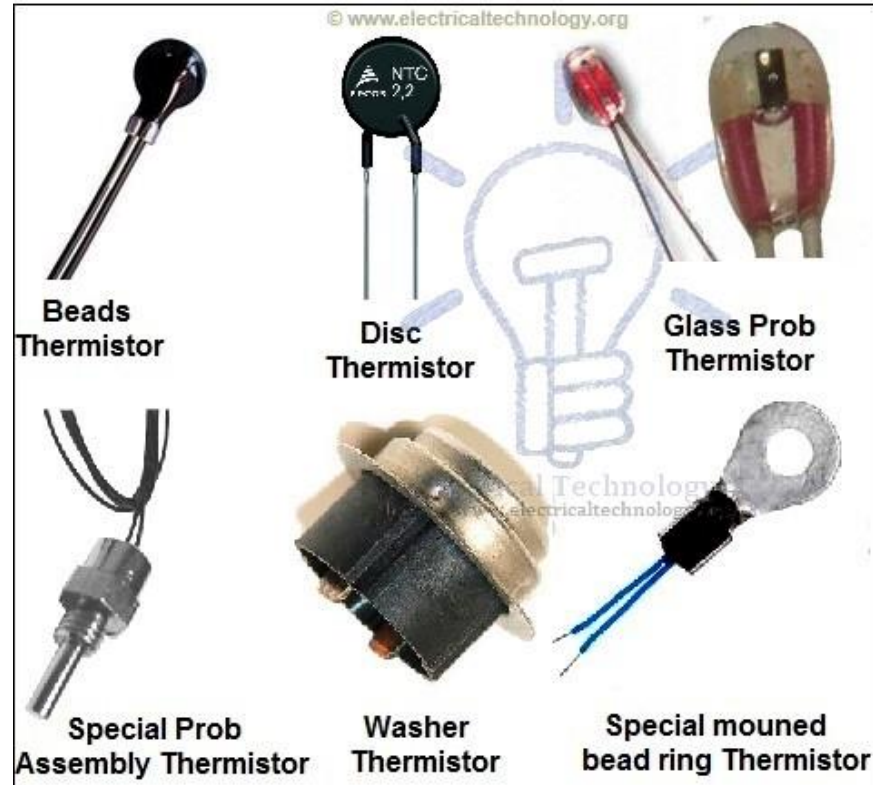
Termistorul

$$R_T = R_0 \cdot e^{\beta \left(\frac{T_0 - T}{T_0 \cdot T} \right)}$$



SV
C Termistor

Termistorul



Fotorezistența

LDR

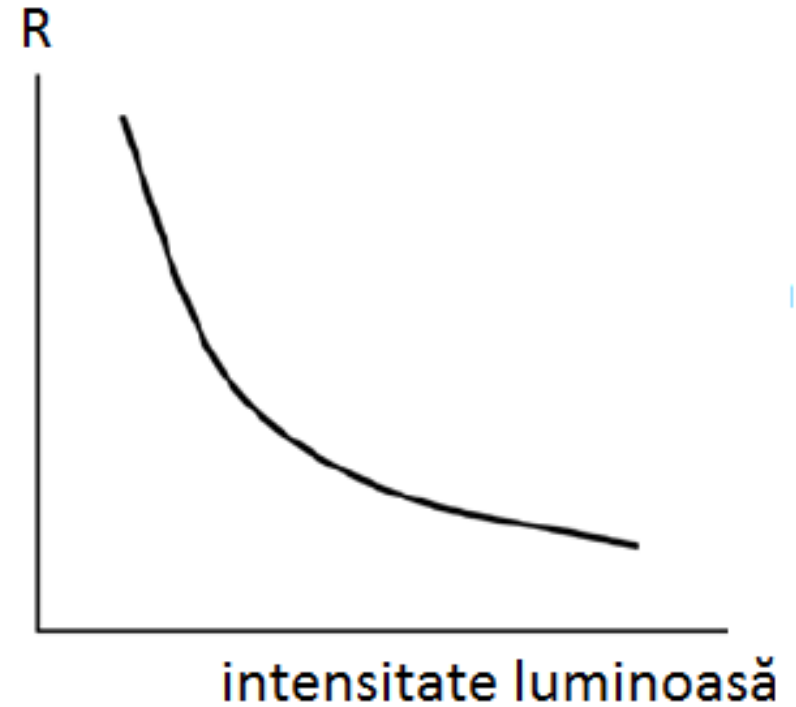
Light-Dependent Resistor

- CdS

- 0,6 μ m
- 1M Ω -întuneric $\leftrightarrow$ 1k Ω -lumină

- CdSe

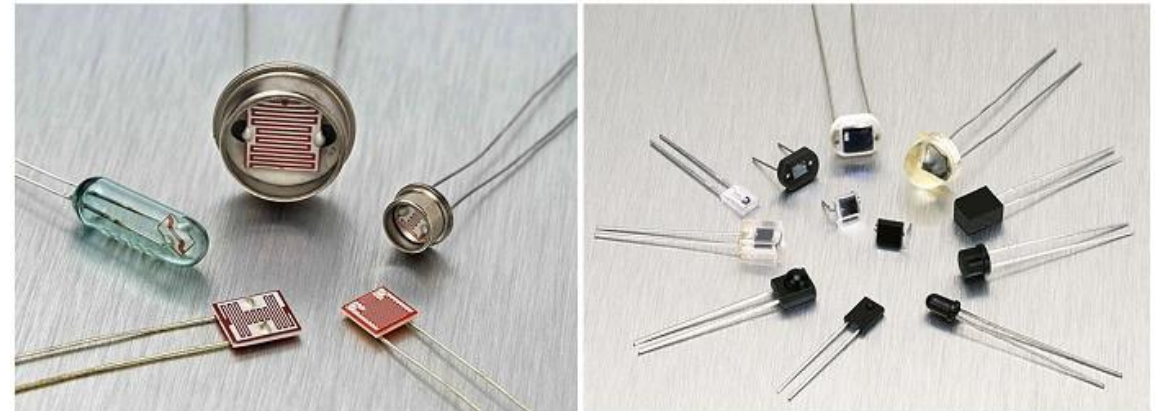
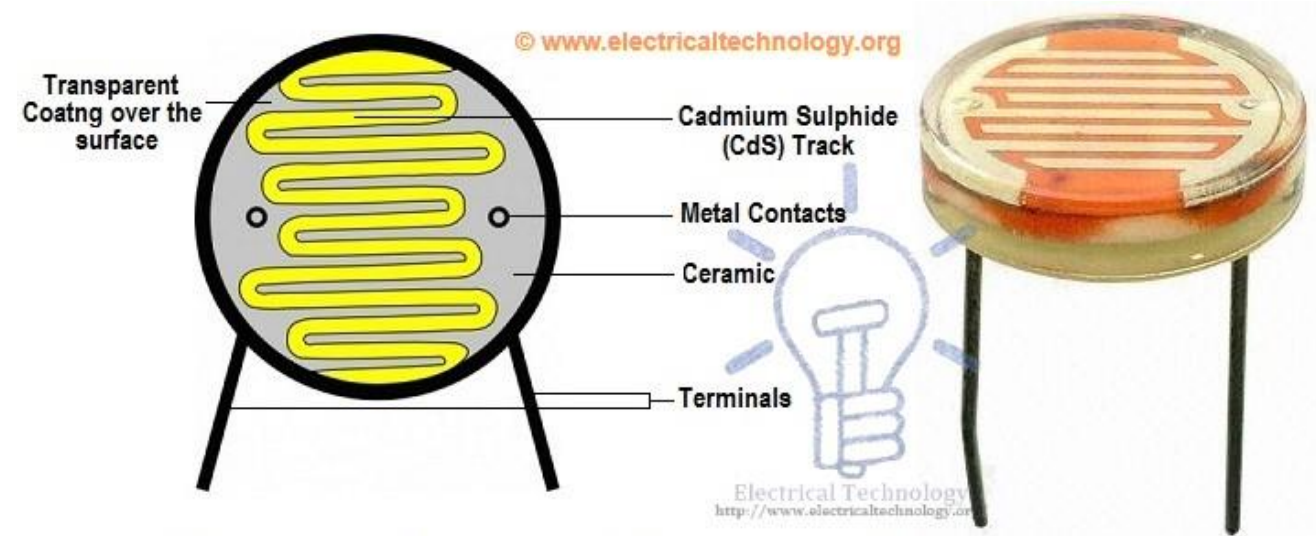
- 0,75 μ m



Fotorezistența

LDR

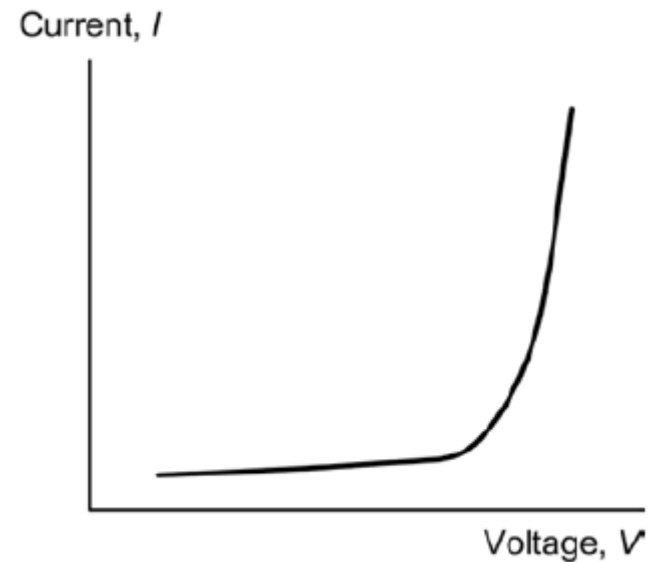
Light-Dependent Resistor



Varistorul

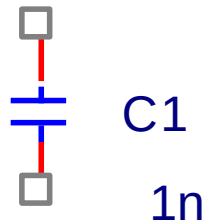
VDR

Voltage-Dependent Resistor

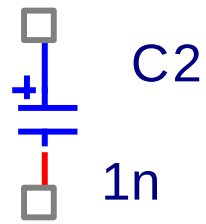


condensatoare

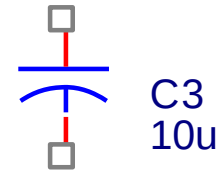
Condensator - simboluri



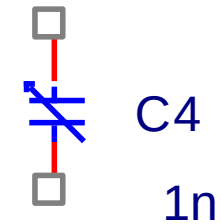
Condensator nepolarizat



Condensatoare polarizate

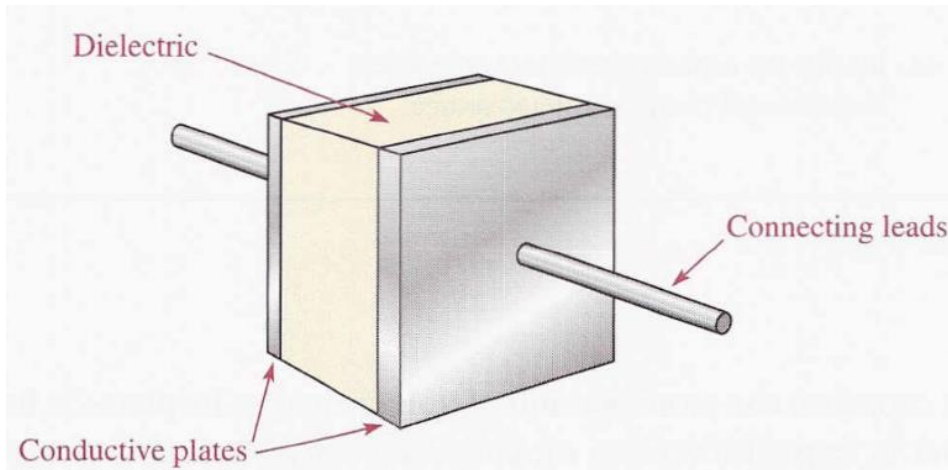


Condensatoar variabil



Condensator - mărimi electrice

$$i_C(t) = C \frac{dv_C}{dt}$$



Sarcina electrică: $Q = C \cdot V$

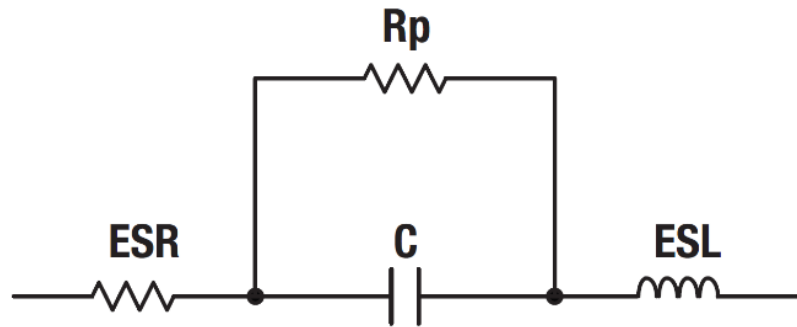
Energia înmagazinată: $W = \frac{C \cdot V^2}{2}$

Dimensiuni fizice: $C = \frac{\epsilon_0 \epsilon_r A}{d}$

Condensator – parametri

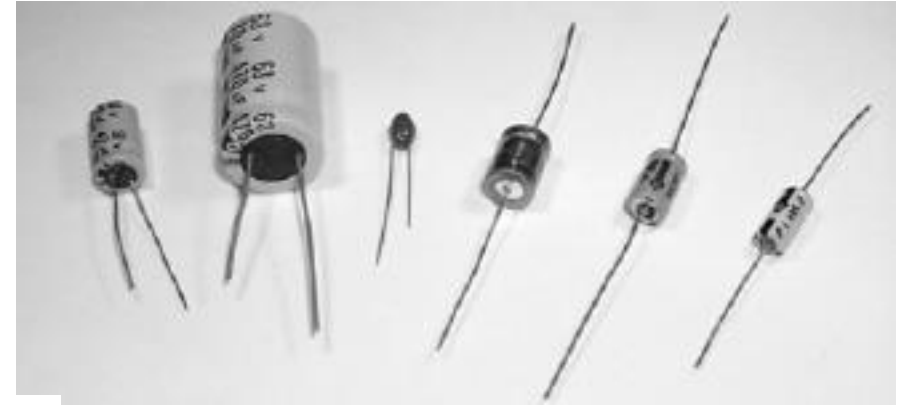
Proprietate	Ceramică	Electrolit	Poliester	Mica	Polistiren
Capacitate [F]	2,2p÷100n	100n÷10m	10n÷2,2μ	0,47÷22k	10p÷22n
Toleranță [%]	±10, ±20	-10÷+50	±10	±1	±5
Tensiune [V]	50÷200	6,3÷400	100÷400	350	100
Coeficient de temperatură [ppm/°C]	÷100÷-4700	+1000	+100÷+200	+50	+250
Stabilitate	Acceptabilă	Scăzută	Bună	Excelentă	Bună
Temperatura mediului	-85÷+85	-40÷+80	-40÷+100	-40÷+125	-40÷+100
Aplicații	De frecvență înaltă și cost scăzut	Filtrare, decuplare	Generale	Oscilatoare, acord	Generale

Condensator – model electric

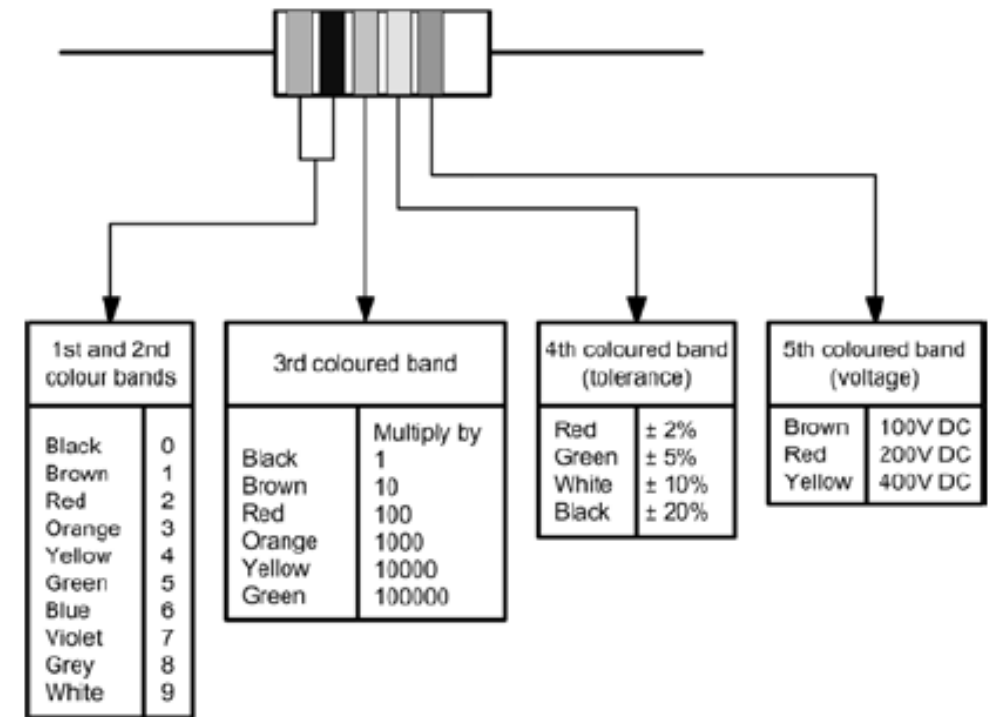
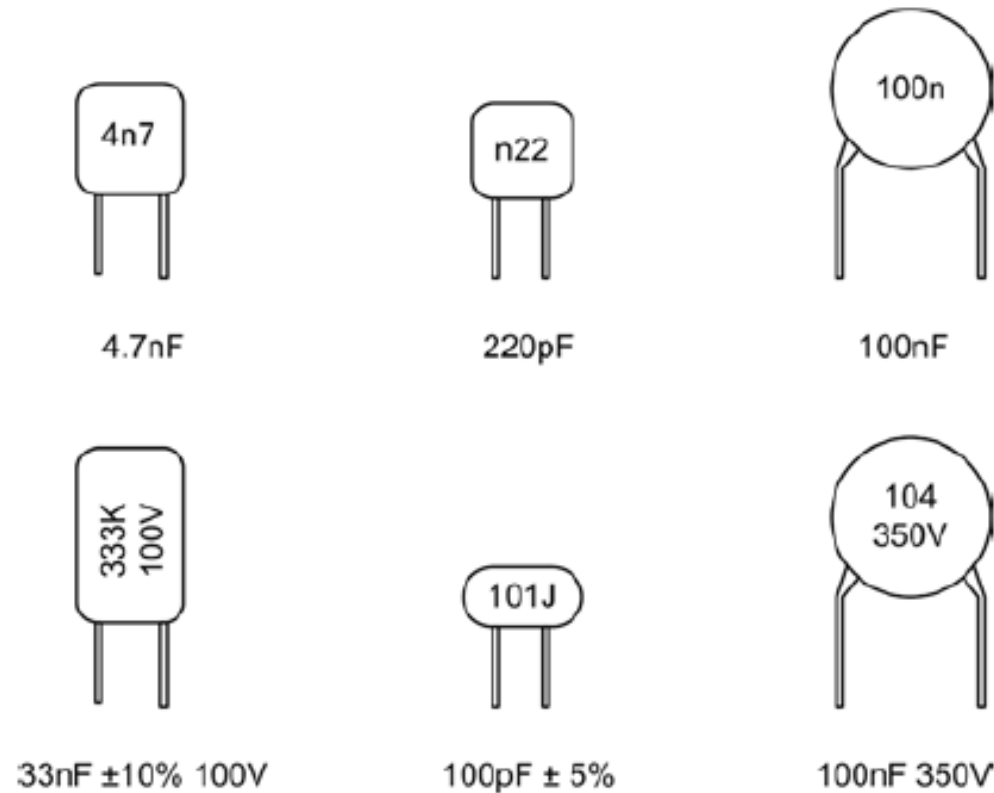


Notăție	Mărime fizică	Valoarea ideală
C	Capacitatea	nominală
ESR	Rezistența serie echivalentă	nulă
ESL	Inductanța serie echivalentă	nulă
Rp	Rezistența paralelă de pierderi	infinită

Condensator



Condensator - marcire

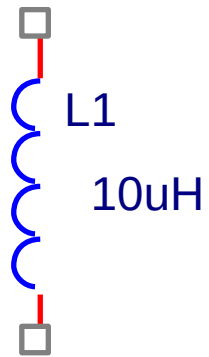


Condensator – aplicații

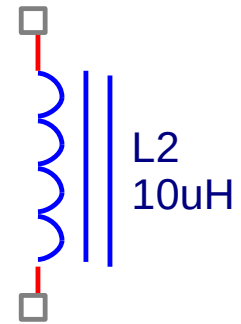
- Stocare de energie
- Temporizare
- Memoria dinamică

bobine

Bobina - simboluri

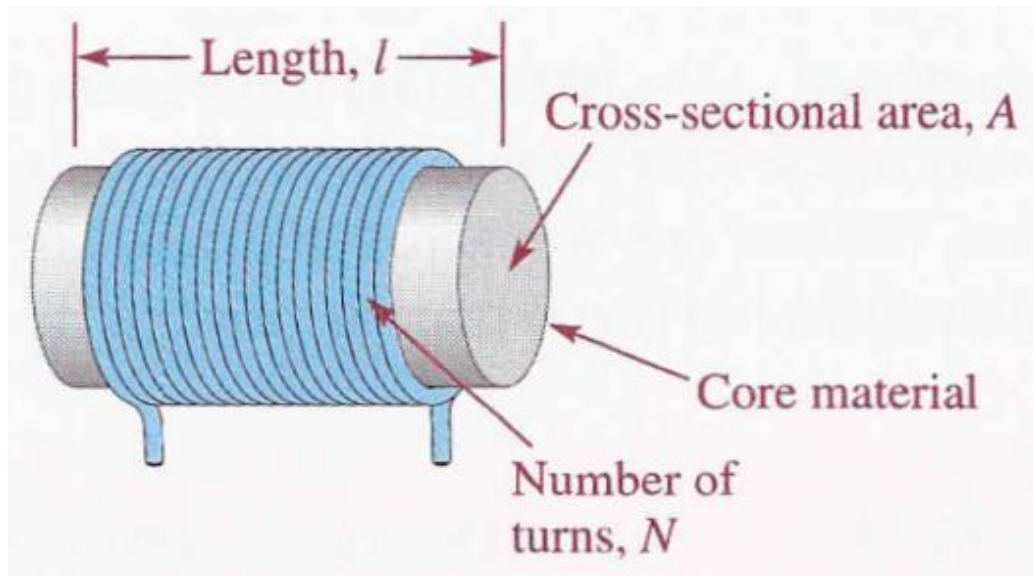


Solenoid fără miez



Solenoid cu miez

Bobina - mărimi electrice



$$V_L = -L \frac{di}{dt}$$

Energia înmagazinată: $W = \frac{L \cdot I^2}{2}$

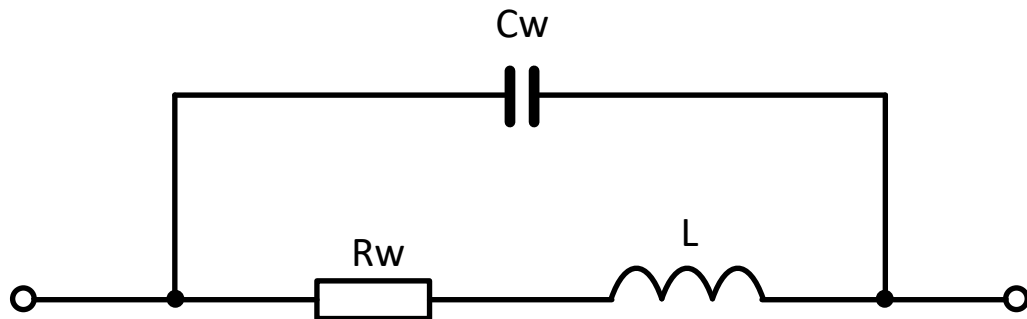
Dimensiuni fizice: $L = \frac{\mu_0 \mu_r N^2 A}{l}$

Bobina – parametri



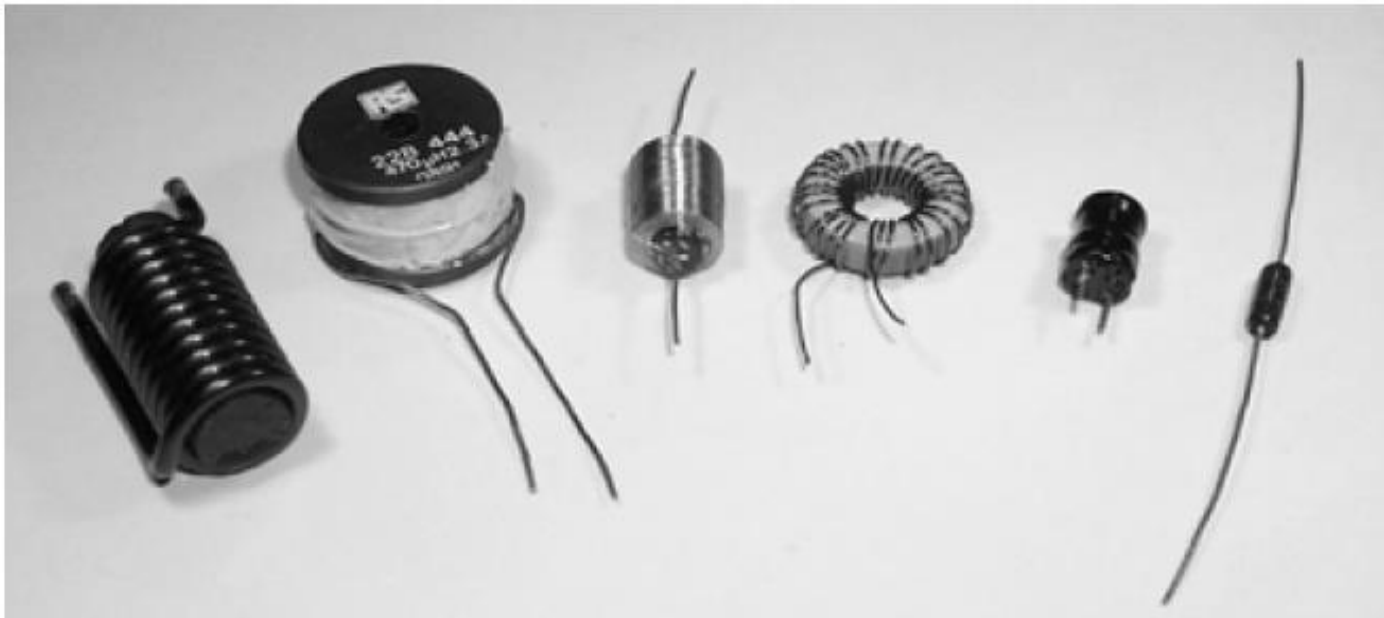
Proprietate	Aer	Miez de ferită	Tor de ferită	Miez de fier
Inductanță [H]	50n÷100μ	10μ÷1m	1m÷100m	20m÷20
Toleranță [%]	±5	+10	±10	±20
Rezistență de cc [Ω]	0,05÷5	0,1÷10	5÷100	10÷200
Factor de calitate tipic, Q	60	80	40	20
Domeniu de frecvență [Hz]	1MHz÷500MHz	100kHz÷100MHz	1kHz÷10MHz	50Hz÷10kHz
Aplicații	Filtre, circuite de acord	Filtre, transformatoare HF	Filtre și transformatoare LF și MF	Filtre, bobine șoc

Bobina – model electric

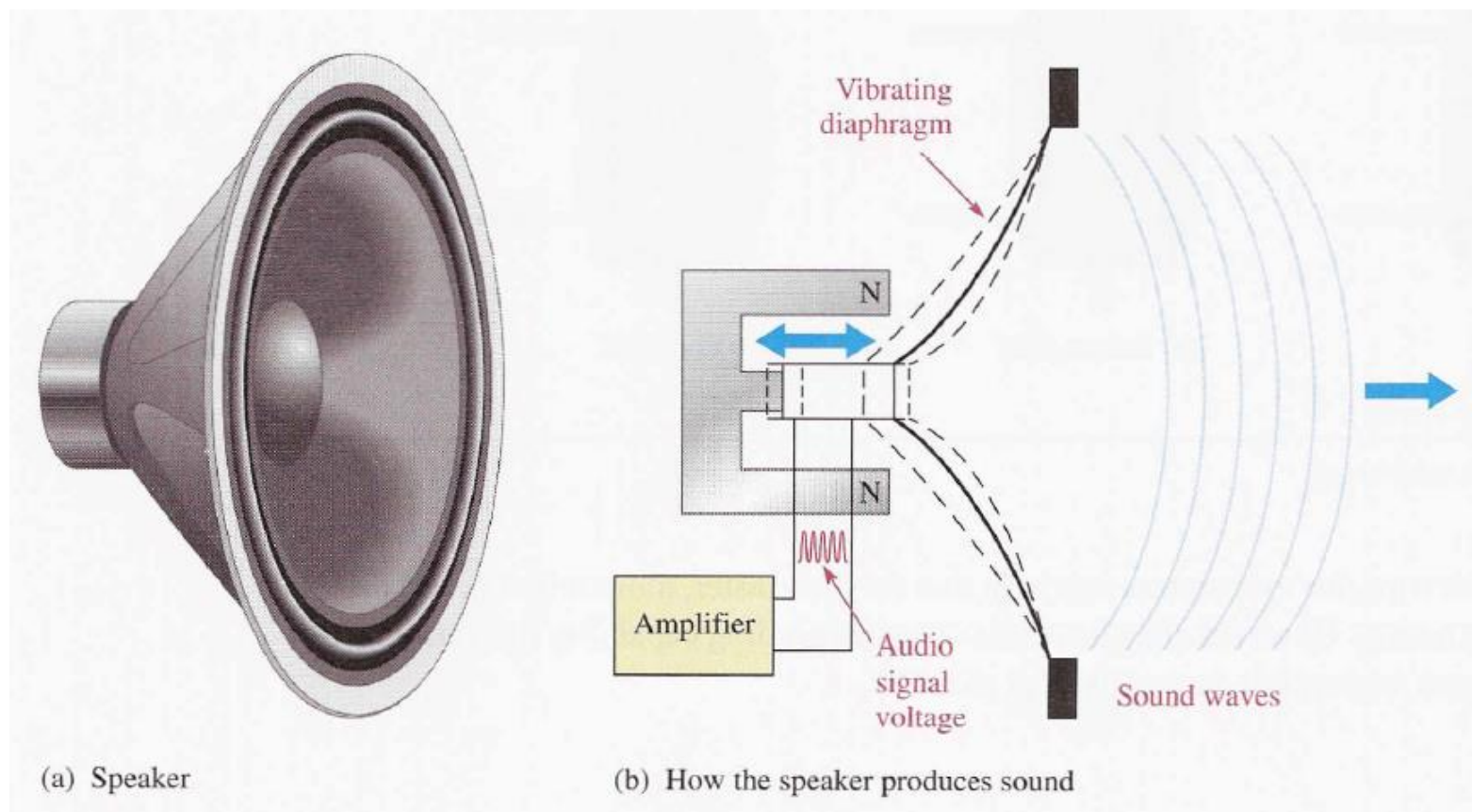


Notăție	Mărimă fizică	Valoarea ideală
L	Inductanța	nominală
C_w	Capacitatea paralelă parazită	nulă
R_w	Rezistența serie de pierderi	nulă

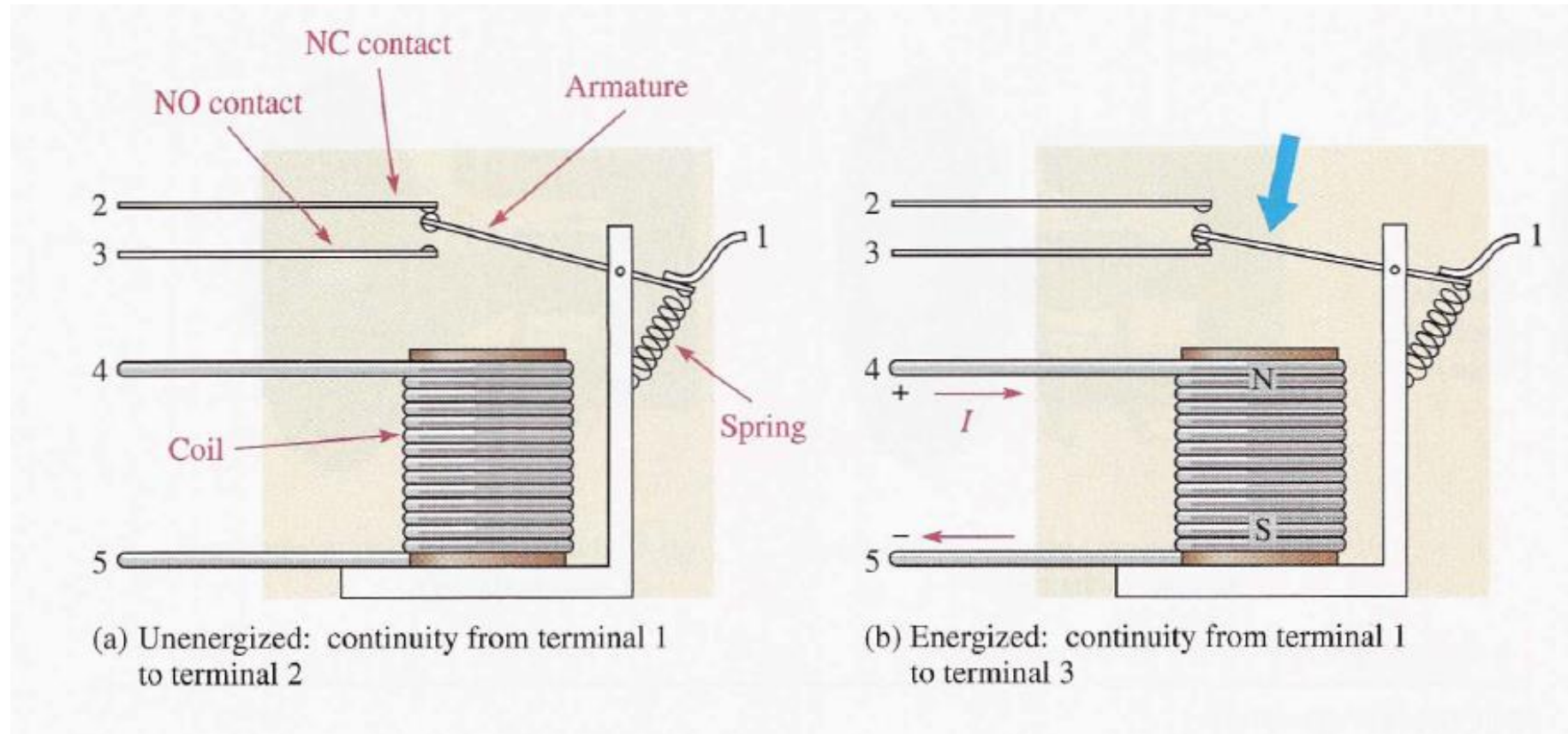
Bobina



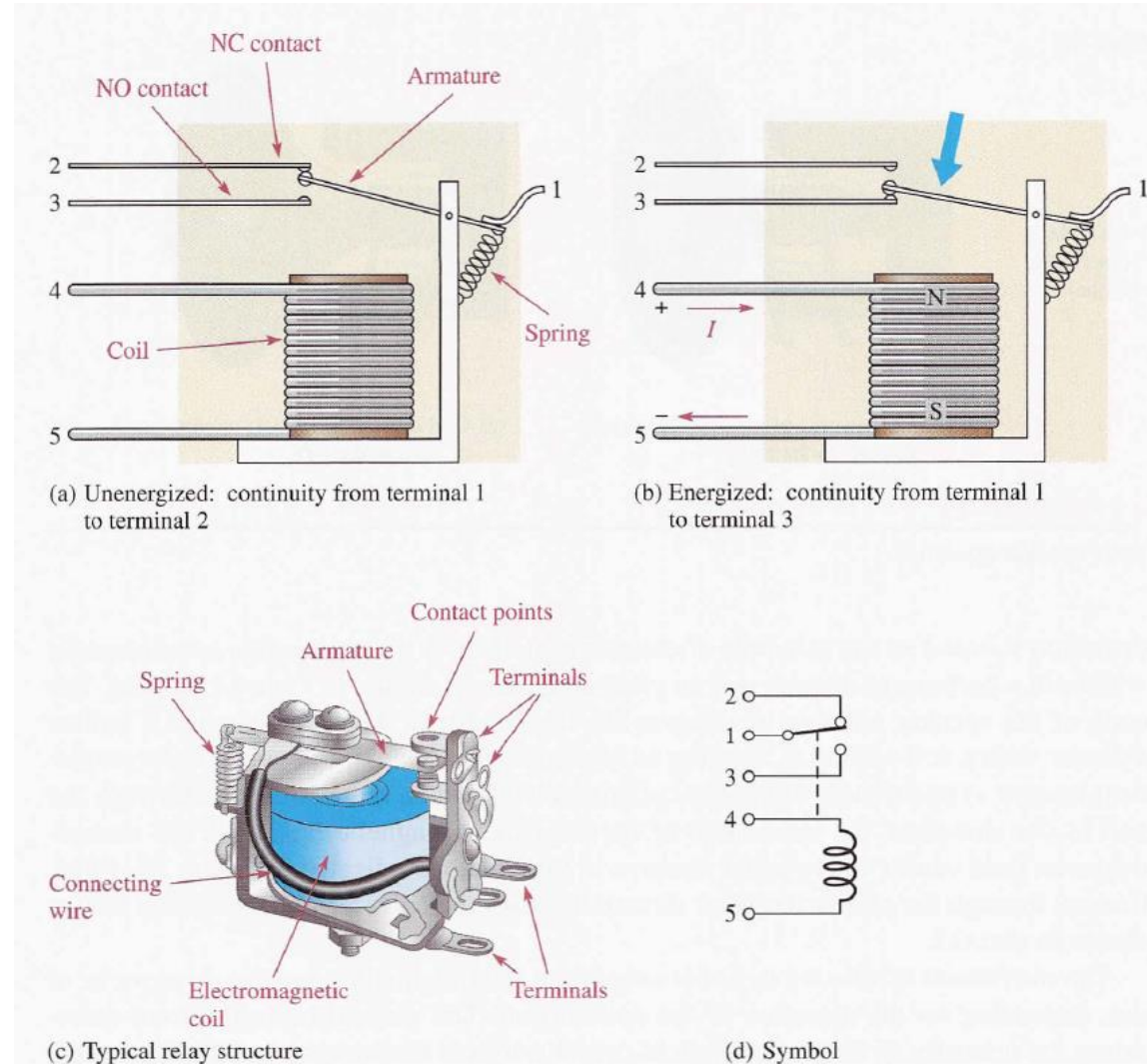
Bobina – aplicații



Bobina – aplicații

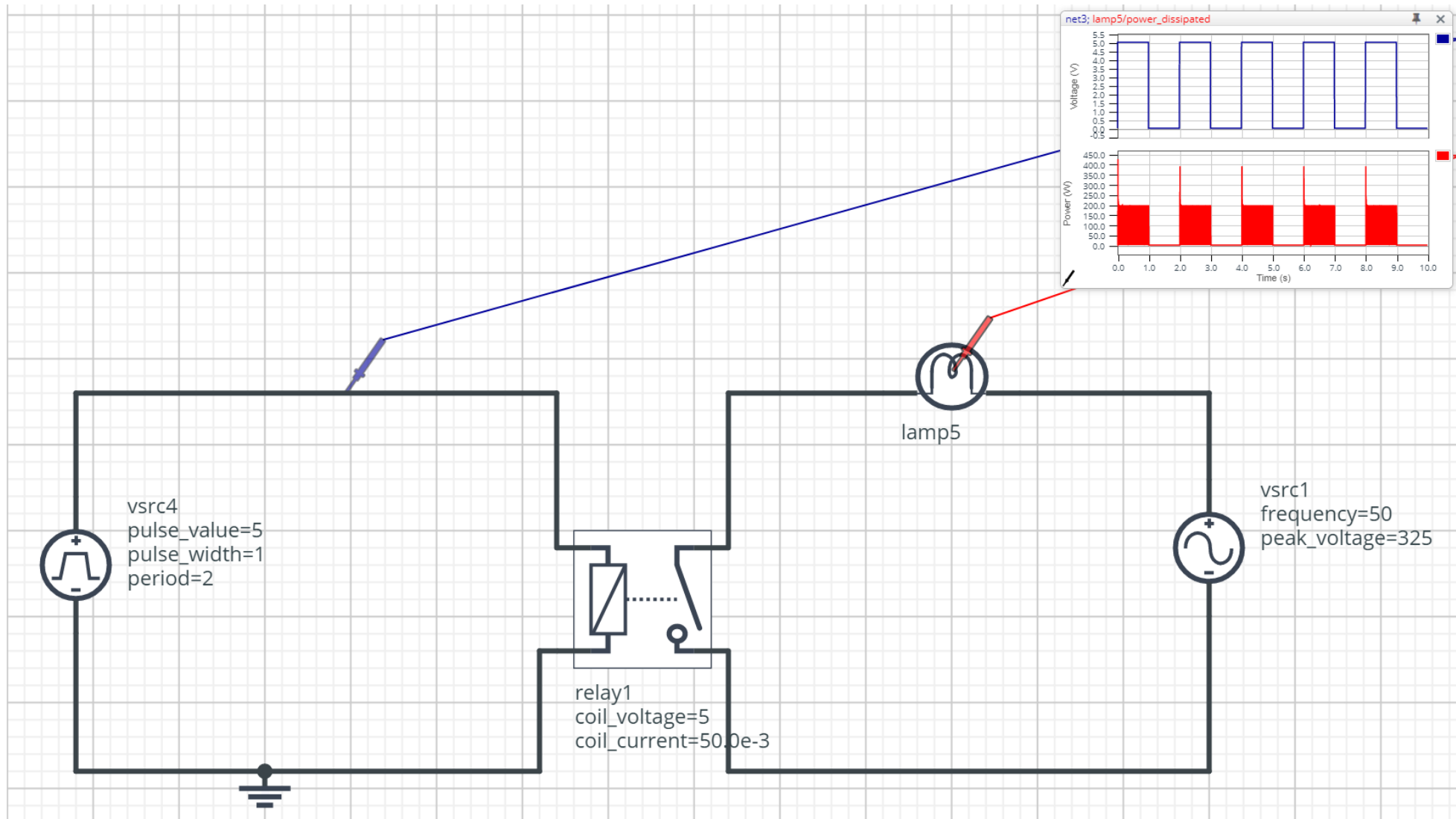


Bobina – aplicații



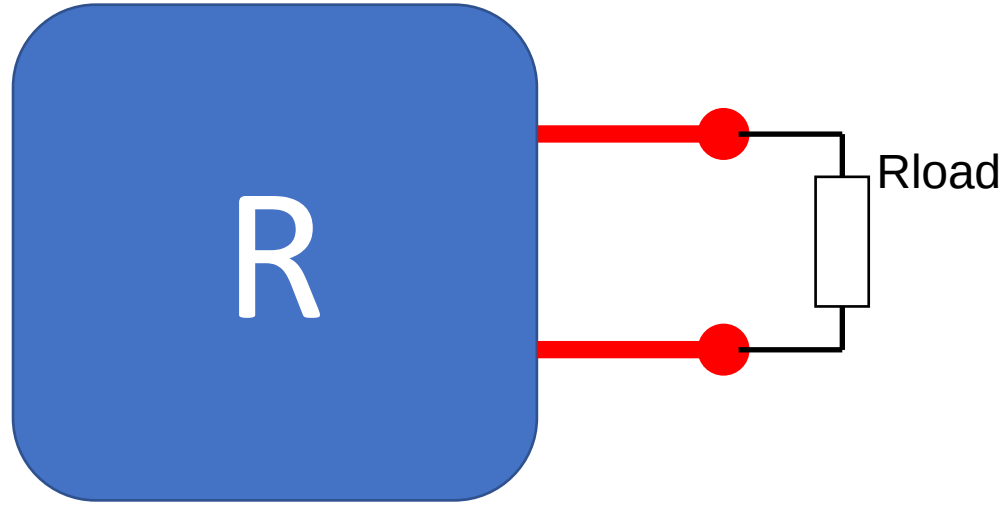
Bobina – aplicații

SV
C Releu electromagnetic

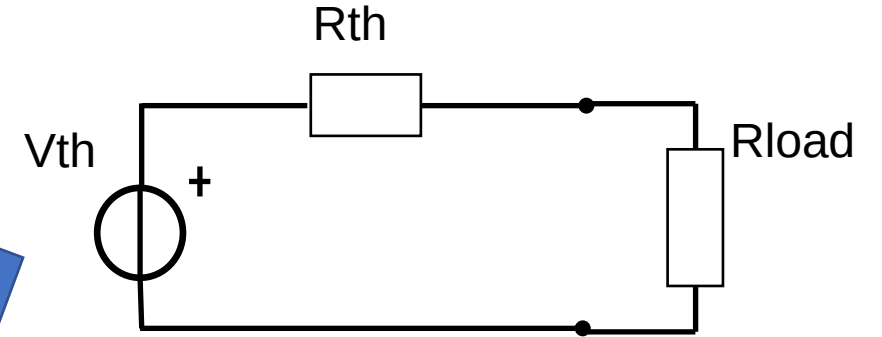


teoreme

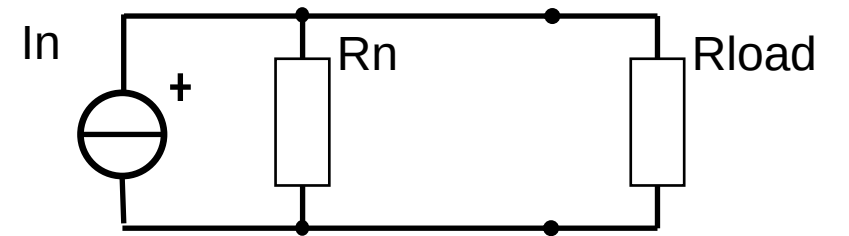
teoreme



Thévenin



Norton



teoreme

Léon Charles **THÉVENIN**

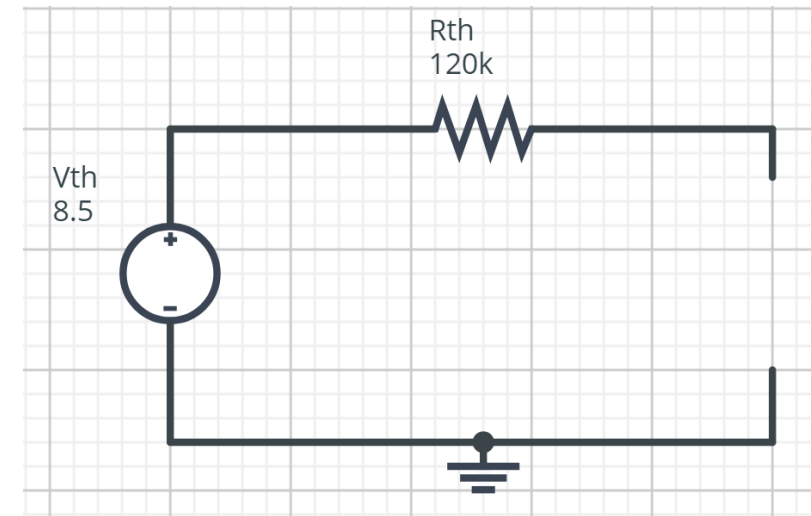
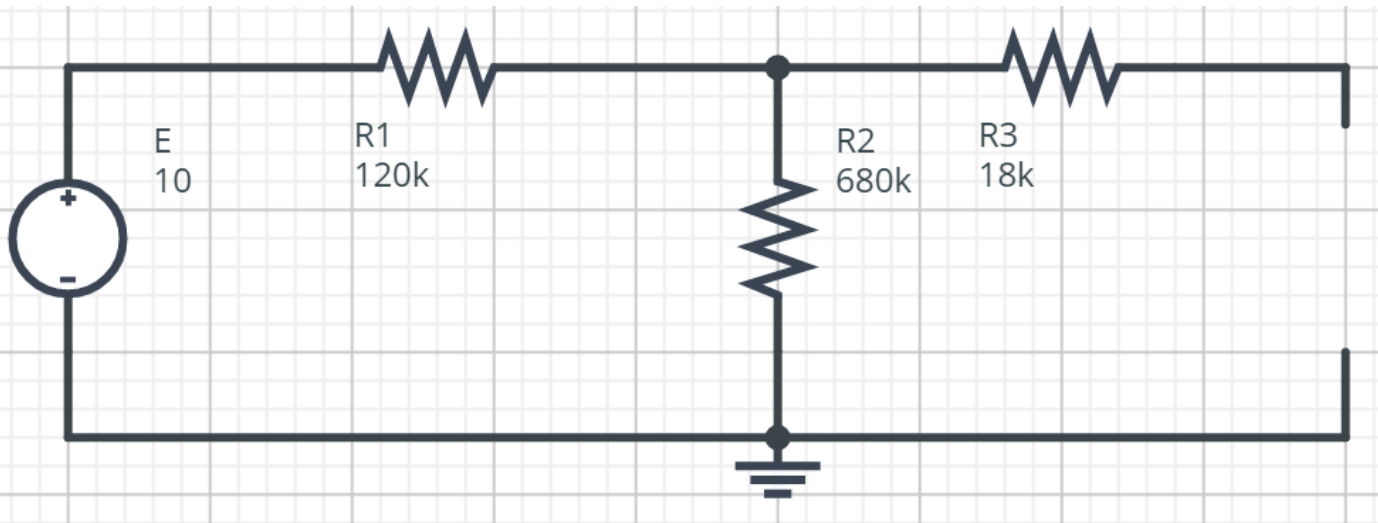
1857-1926



Léon Charles Thévenin

1857–1926

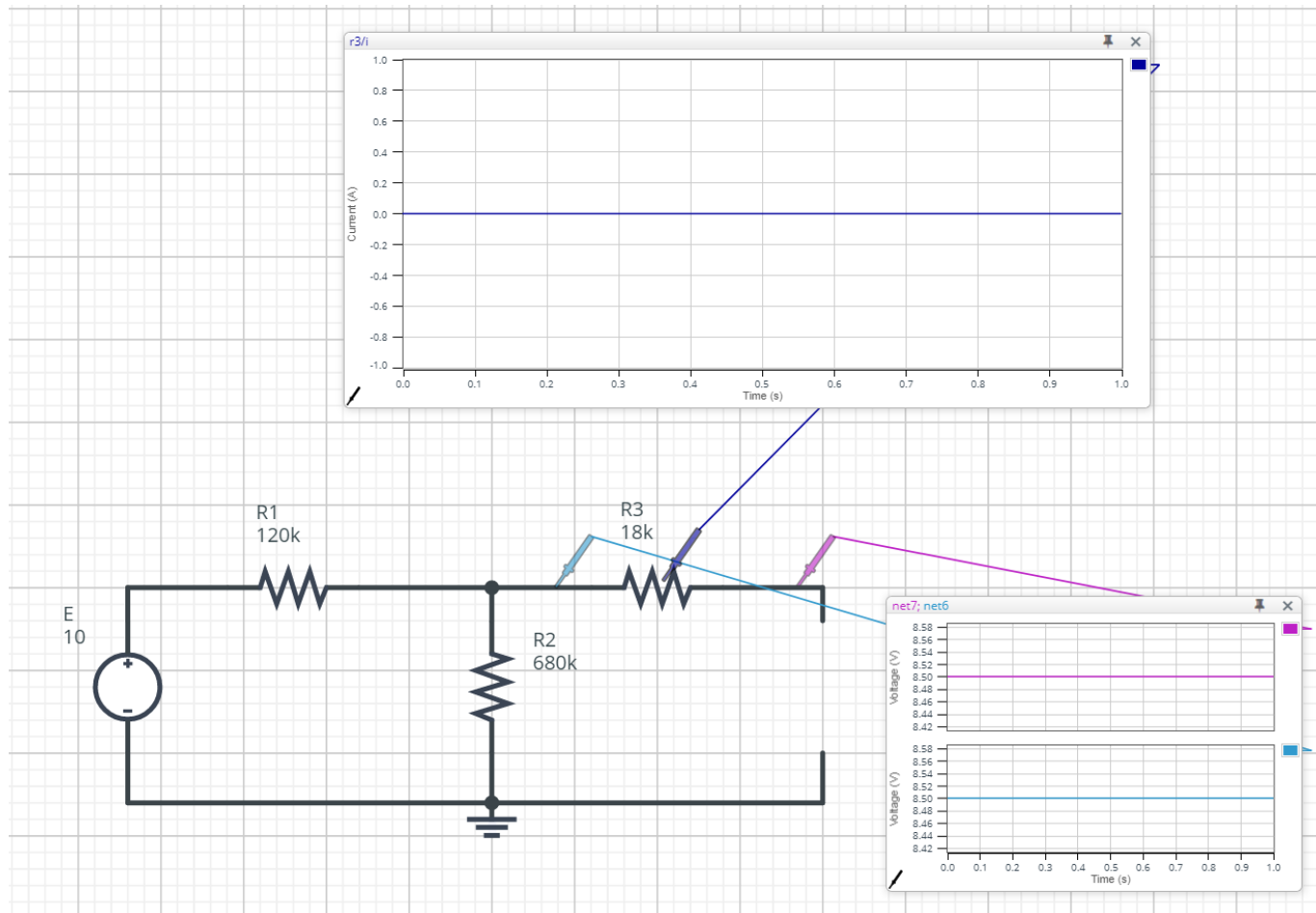
THÉVENIN



THÉVENIN

V_{th} = tensiunea la mers în gol

SV
C Thevenin

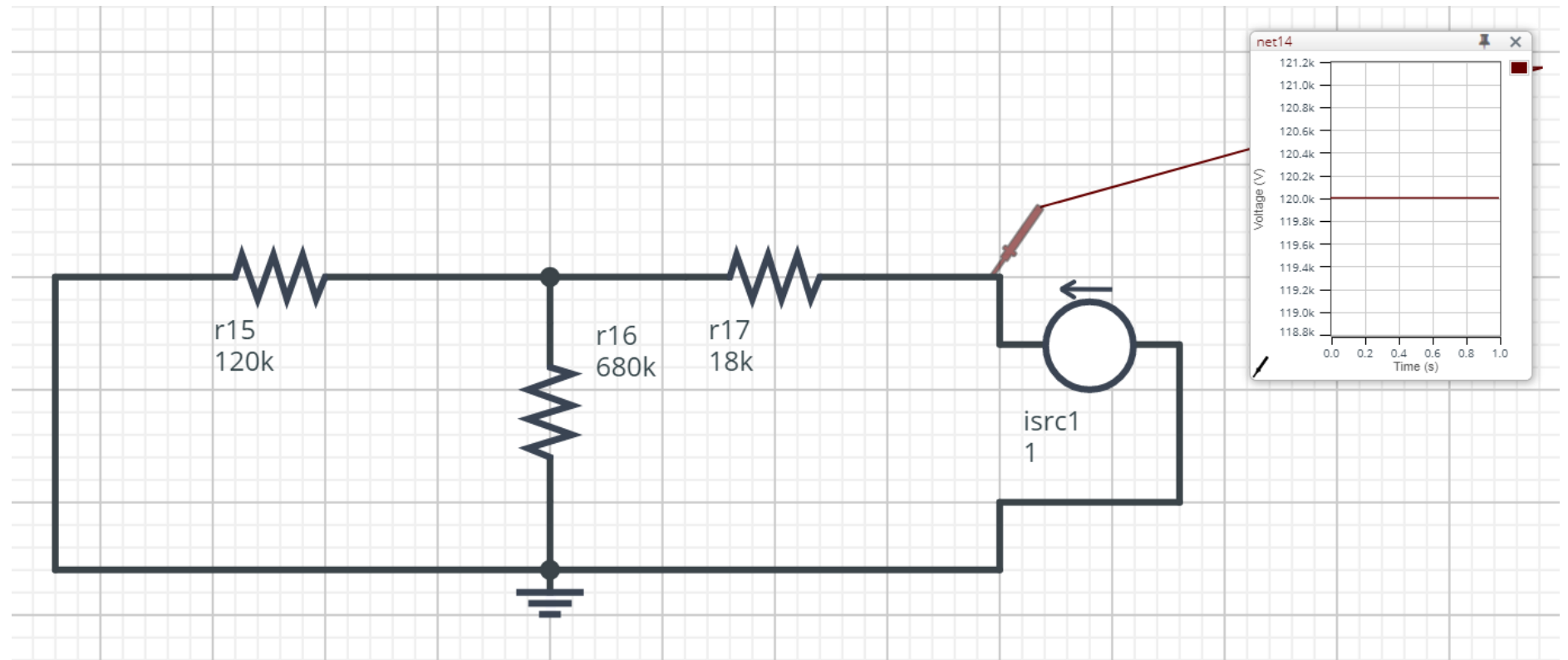


THÉVENIN

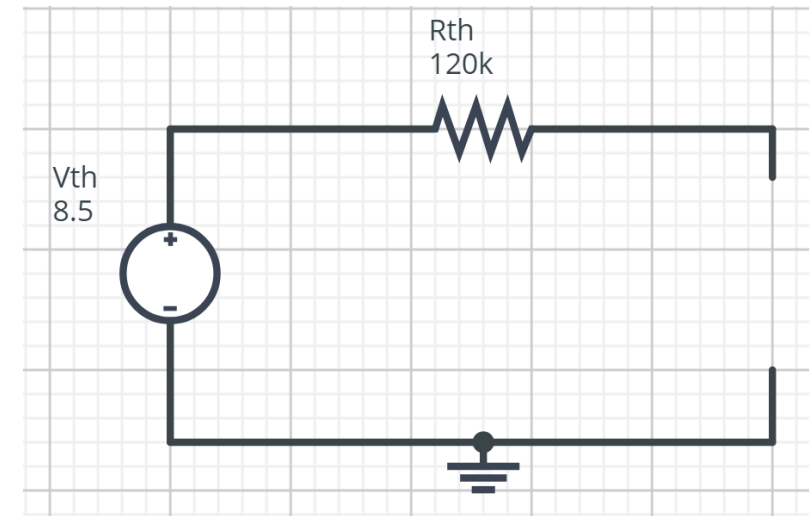
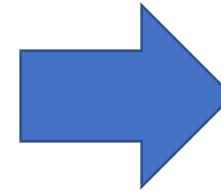
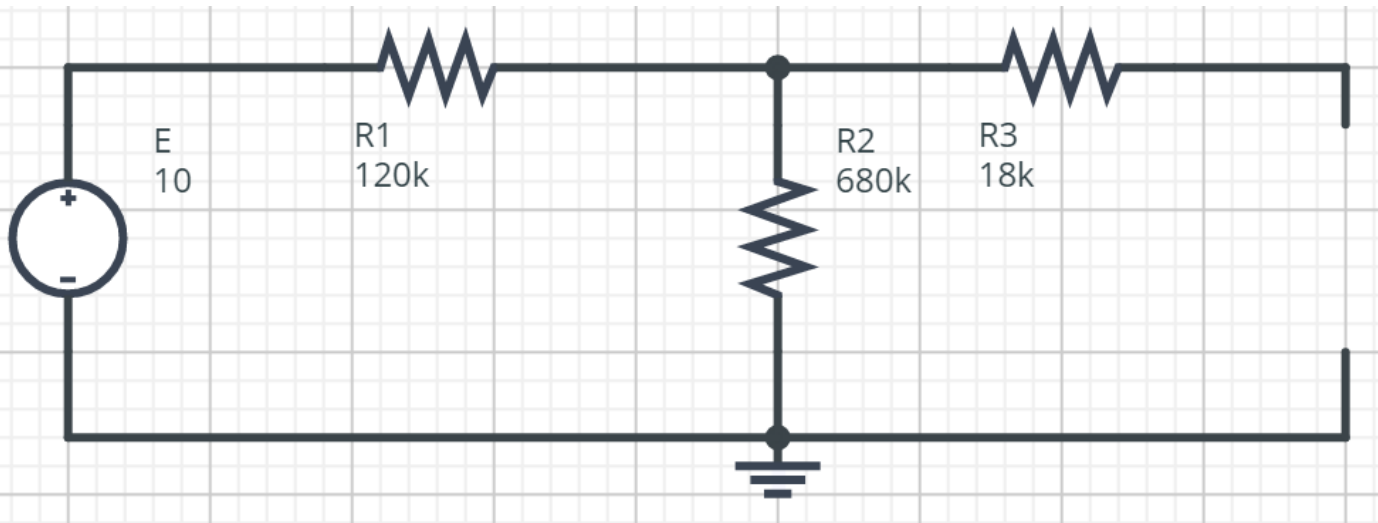
R_{th} = rezistența echivalentă

- la mers în gol
- după pasivizare

SV C Thevenin



THÉVENIN



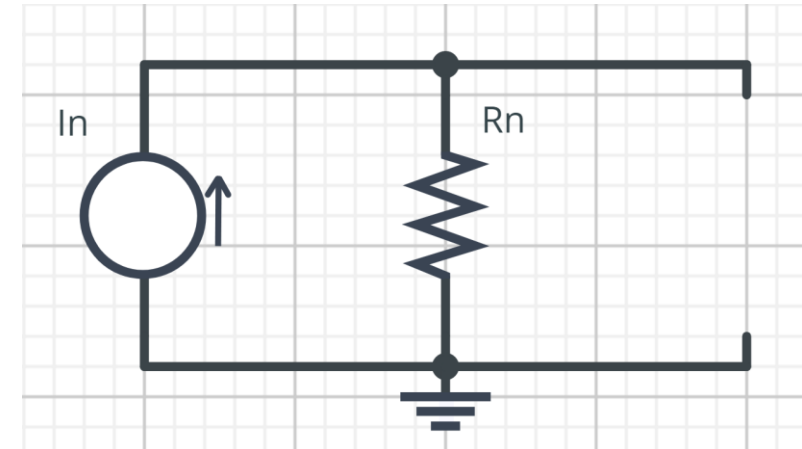
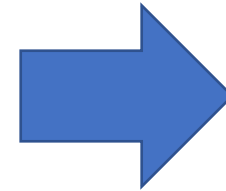
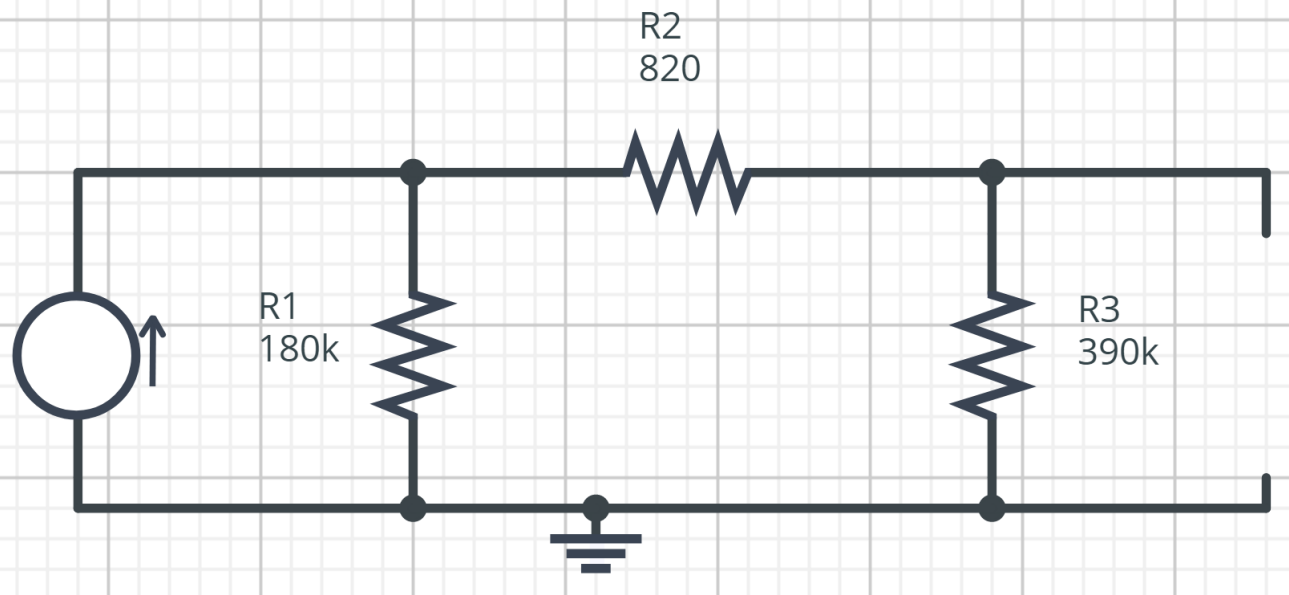
teoreme

Edward Lawry **NORTON**

1898-1983

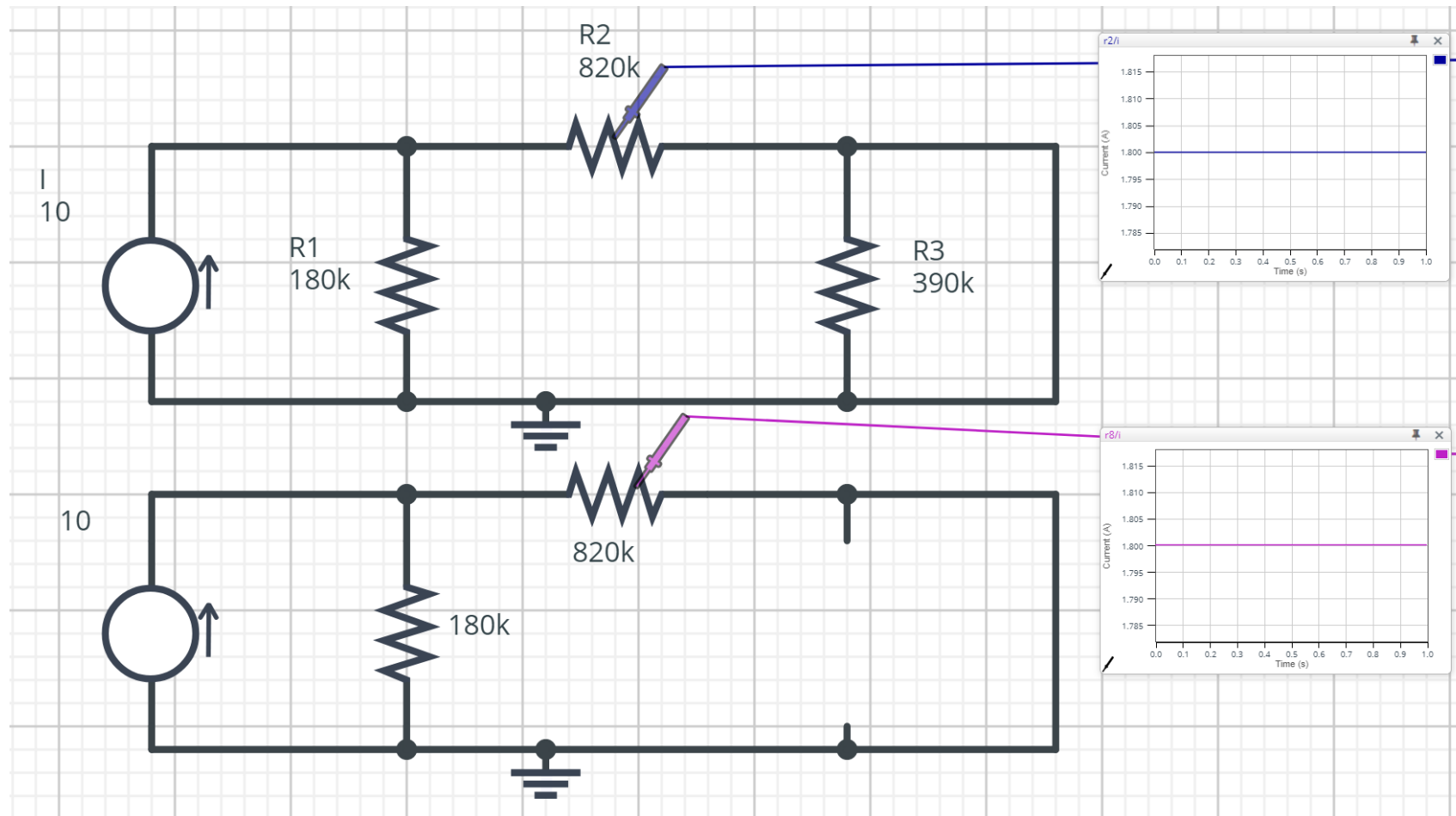


NORTON



NORTON

I_n = curentul la scurt-circuit

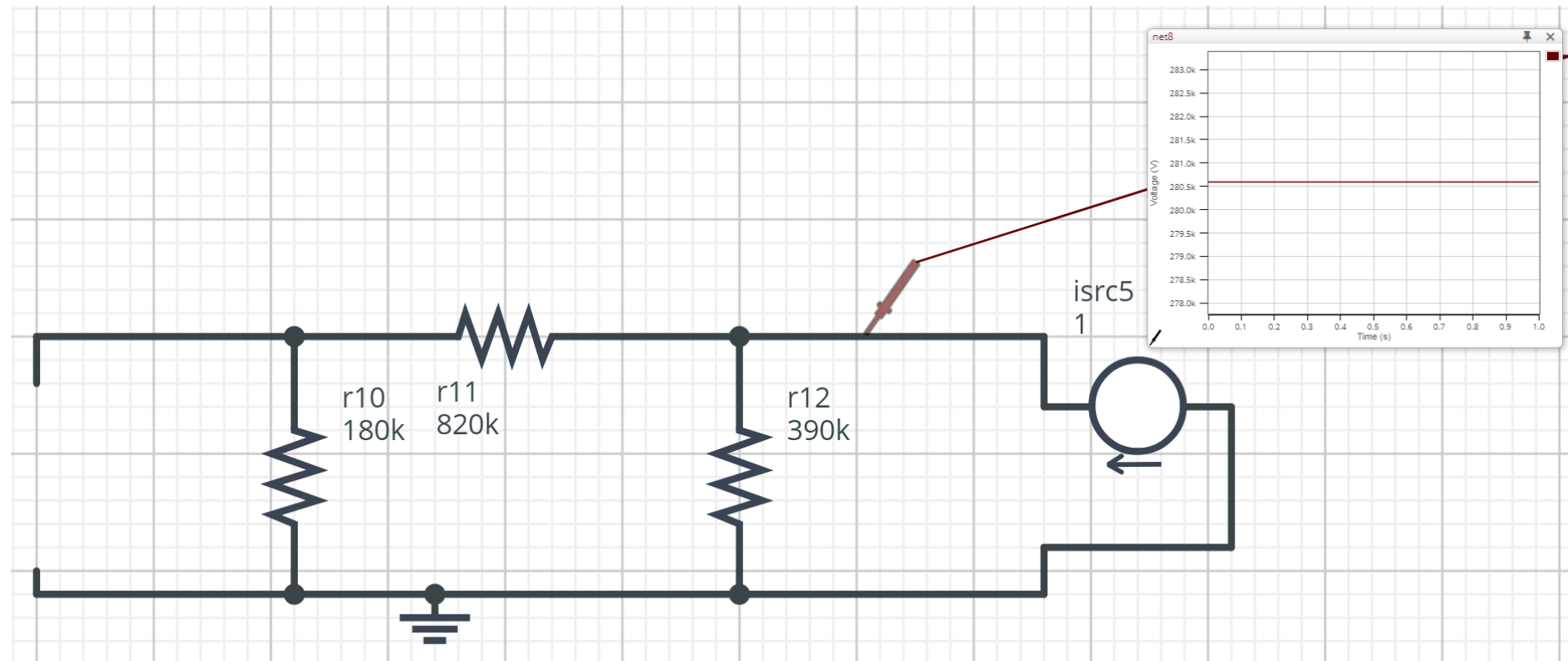


NORTON

R_n = rezistența echivalentă

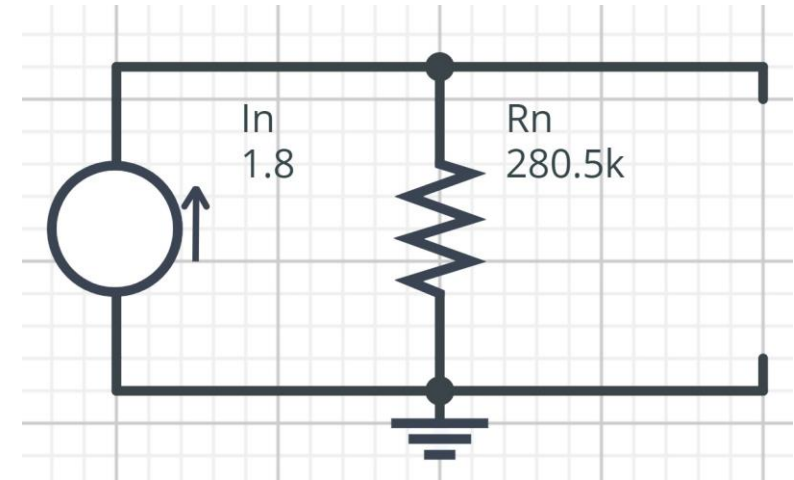
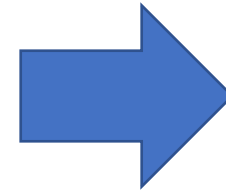
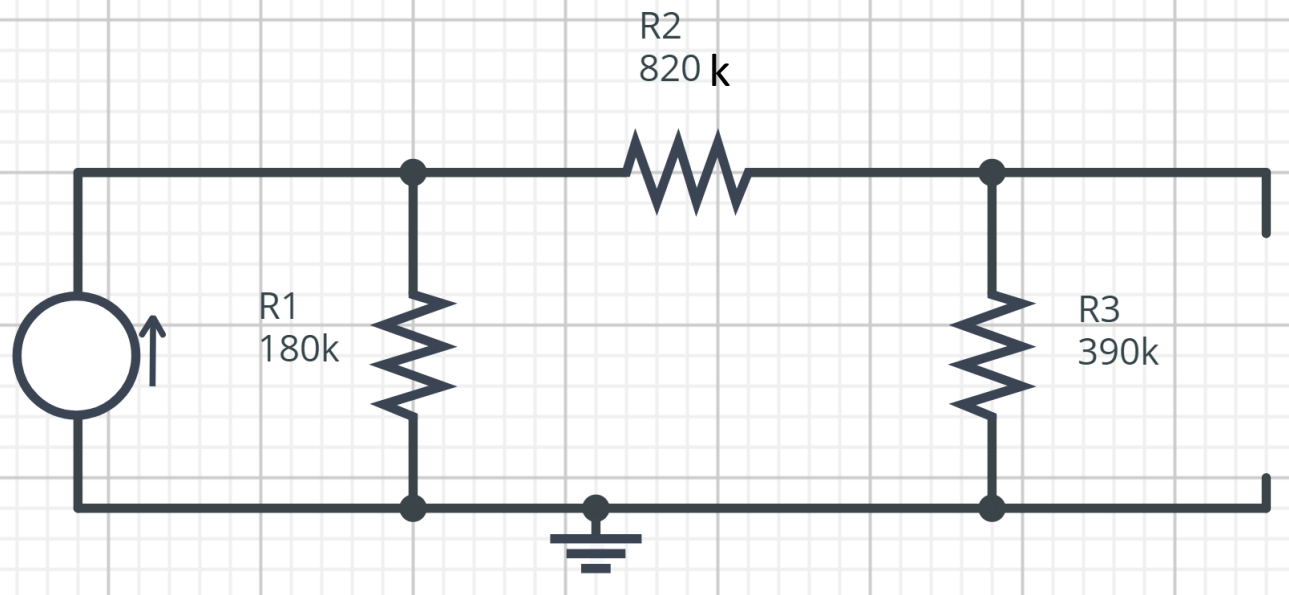
- la mers în gol
- după pasivizare

SV
C Norton



NORTON

SV
C Norton

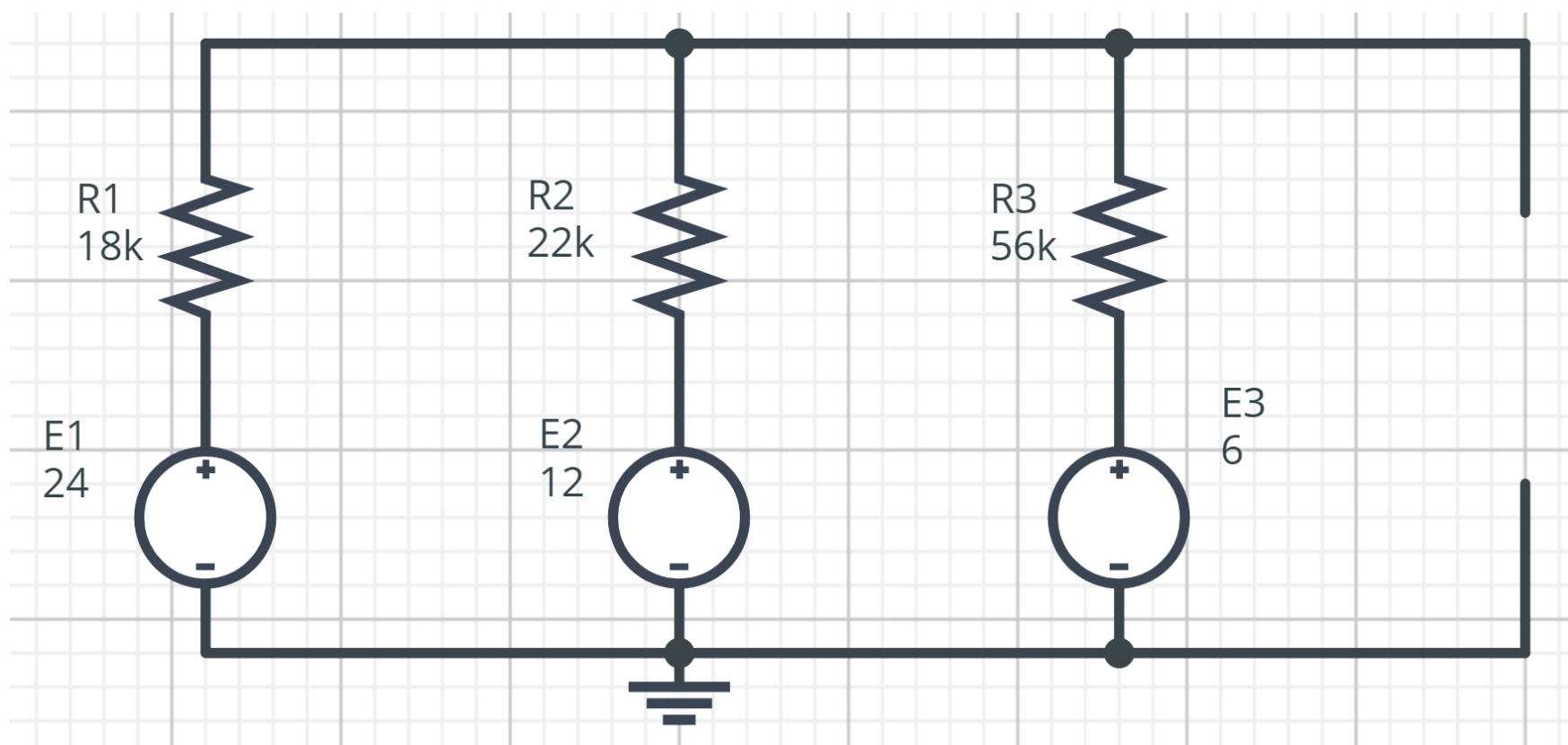


Suprapunerea efectelor

Efectul produs la bornele de ieșire ale unei rețele pur rezistive este egal cu suma algebrică a efectelor parțiale calculate pentru fiecare sursă în parte în situația în care toate celelalte surse sunt pasivizate.

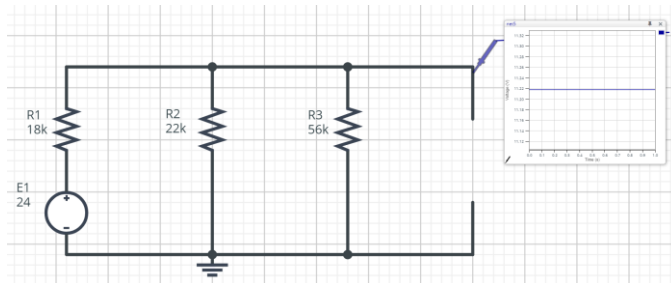
Suprapunerea efectelor

SV
C Suprapunerea efectelor

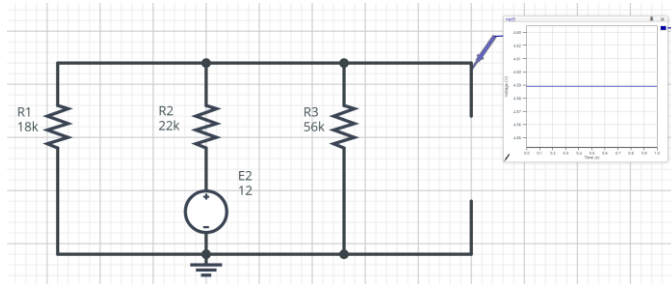


Suprapunerea efectelor

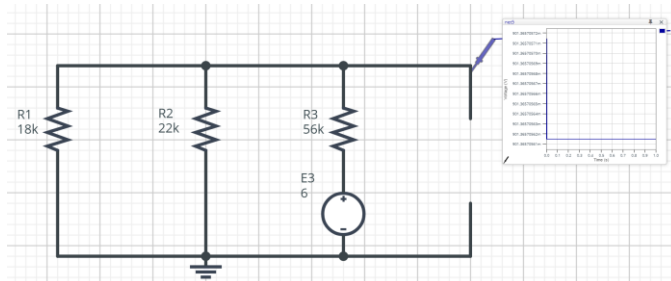
SV
C Suprapunerea efectelor



$$V_{o1} = 11,2V$$

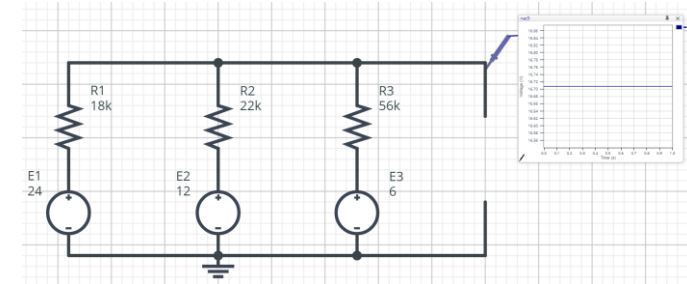


$$V_{o2} = 4,6V$$



$$V_{o3} = 0,9V$$

$$V_o = V_{o1} + V_{o2} + V_{o3} = 16,7V$$



next

- Comportarea în timp a circuitelor RC
- Comportarea în frecvență a circuitelor RC